

ONLINE REVIEWS AND INFORMATION OVERLOAD: THE ROLE OF SELECTIVE, PARSIMONIOUS, AND CONCORDANT TOP REVIEWS¹

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By empowering customers to make fitting purchases, user reviews play an important role in reducing inefficiencies in the provisioning of product information. Because of the abundance of reviews and the signals they provide, this information may become confusing and risks overloading customers. Consequently, review hosting platforms have adjusted their designs to feature a signal “distilled” from a selective set of “top reviews” and their valences. The expected ease with which customers process this signal is intended to increase their satisfaction, thus reducing dispersion in their subsequent review ratings. In this study, we analyze the influential role that top reviews and their valence play under various scenarios: when customers are overloaded by a large number of reviews, when top reviews themselves are not parsimonious in number, and when the signals from top reviews are not in concordance with that from all the other reviews. We find that the valence of top reviews plays a central role in mitigating information overload. However, the influence of those top reviews diminishes when they too pose an overload risk but is strengthened when their signal is reaffirmed by signals from all other reviews. Finally, the impact of top reviews is weaker for less popular products.

Keywords: Information overload, top reviews, signaling theory, information theory, information provisioning

Introduction

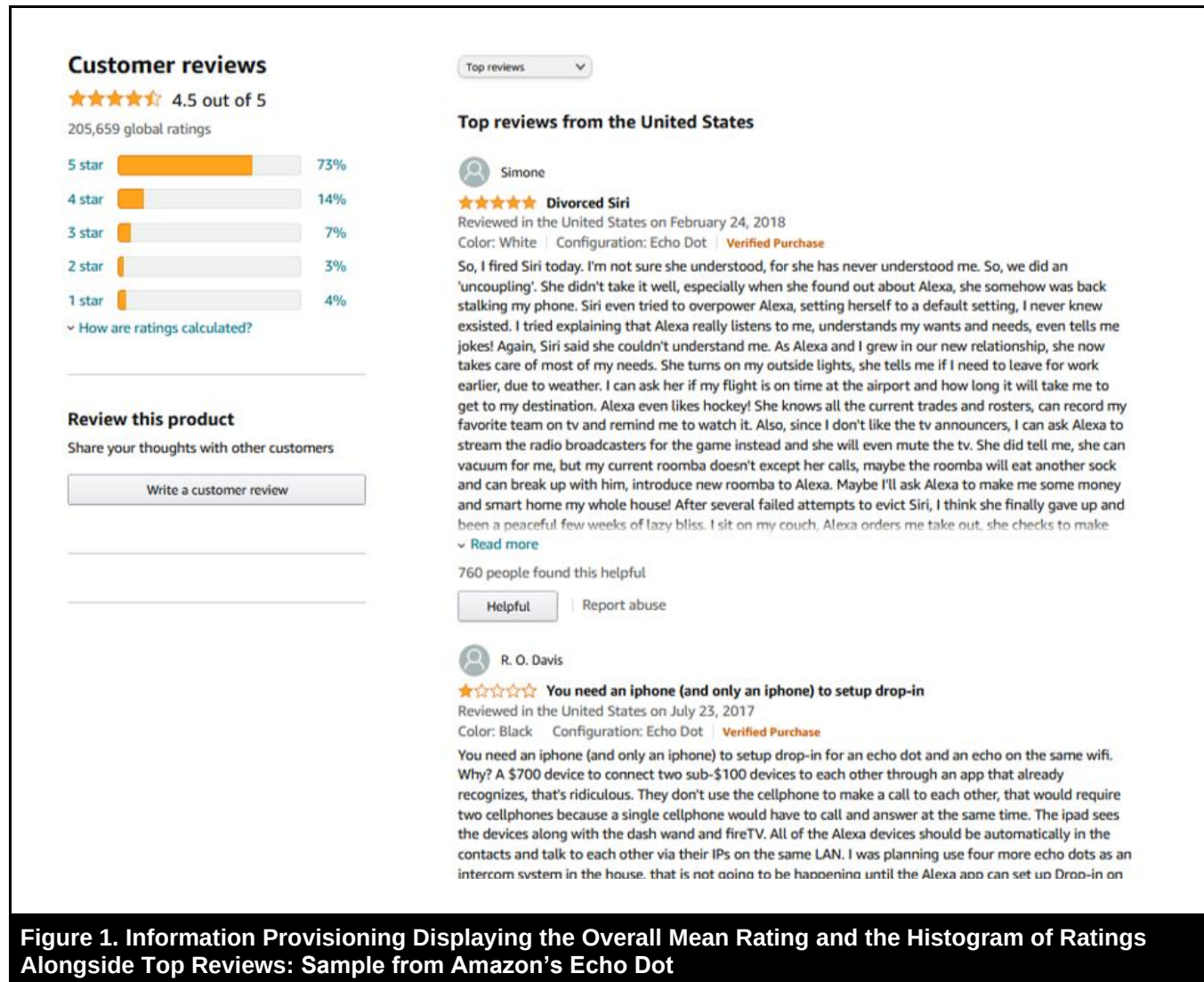
We don't make money when we sell things. We make money when we help customers make better decisions

— Jeff Bezos, Amazon

Online review hosting platforms are well-recognized as information-rich environments where crowd-sourced reviews help reduce the uncertainty surrounding prospective purchases, thus reaffirming the utility of information (Roberts & Urban, 1988). The wealth of available information, however, can be taxing and may overload subsequent customers (Dai et al., 2018; Davis et al., 2021; Jacoby, 1984). Amazon's Echo Dot,

for example, gathered a whopping one million reviews in the first four years of its release. This volume of information exposes customers to noisy and sometimes conflicting signals (Branco et al., 2016), potentially resulting in poor customer choices. In response, platforms have explored ways to provide information more efficiently to ensure the satisfaction of subsequent customers with their purchases. Since customers respond better to simple messages (Jones et al., 2004), especially in visually rich environments, a key approach that review platforms have implemented is signal selectivity. With this approach, the platform selects a set of “top reviews” that the crowd has already endorsed and displays these prominently (see Figure 1 in the context of Amazon).

¹ Bin Gu was the accepting senior editor for this paper. Jui Ramaprasad served as the associate editor.



As a result, customers can readily view these top reviews and easily deduce their overall valence. The hosting platform's aim is to ensure that the informational signal from the valence of these reviews helps the customer make a satisfactory choice. In rich informational environments with a large number of reviews posted to a product, this role in improving the efficiency of information provisioning is undoubtedly further accentuated. Yet this signal, which derives its value from its selectivity and parsimony, may itself become overwhelming when too many such top reviews are featured at once, rendering it less effective. Additionally, this signal's role may be further diminished when the top reviews are not in concordance with the set of all available reviews. The (dis)agreement is readily apparent to customers, given the platform's well-established practice of displaying a histogram of ratings for all available reviews. The worry is, as Davis et

al. (2021, p. 2928) puts it, that the "benefits of one type of (social information) may effectively be cancelled or crowded out by the other type." This may be the case because signals in concordance are more reliable, simplifying decision-making, and improving the efficiency of information provisioning.

Given the potentially pivotal role played by top reviews and their valences, and by review-hosting platforms in provisioning such signals, it is essential for researchers and practitioners alike to have a systematic understanding of the implications of signals from the valence of top reviews on the satisfaction of subsequent customers, specifically in information-rich environments. The objective of this paper is, therefore, to unravel the role of top reviews and their valences in shaping the efficiency of information provisioning.

Specifically, we study the role they play in the presence of increasing information overload, the role of their parsimony, and the role of their concordance with other reviews. Conceptually, in an efficient information provisioning setting, “the item [that is traded] is allocated to the agent [i.e., the online customer in our setting] that values it most” (Aperjis & Johari, 2010). Thus, there would be very few reasons for satisfied online customers to post new reviews whose ratings deviate substantially from one another. In other words, when prepurchase expectations are formed on the basis of ample information, they should match the postpurchase evaluation, thus resulting in very small levels of disagreement, if any, among subsequent opinions about the product. This overall efficiency of information provisioning, however, has not yet been methodically evaluated. Such an analysis would provide insights into the effectiveness of the industry’s pivot to top reviews as a selective signal in the presence of information overload, the implications of its parsimony, and the concordance with other salient signals. By so doing, we explicitly recognize that the signal from top reviews is intricately intertwined with other review-based signals such as “all reviews” in having an impact on customer satisfaction and their subsequent ratings.

To this end, and given the information overload that customers are subjected to, we develop a research framework rooted in both signaling and information theories to analyze the role that signal selectivity, parsimony, and concordance play in shaping the efficiency of information provisioning. The framework highlights *top reviews* and their valences as a major source of information with which customers are presented and highlights their relationship to *all reviews*. The signal from all reviews in the form of the mean rating is widely known to be impactful, thus naturally serving as a strong signal about the product. Yet as more reviews arrive, the informational signal it provides may get cluttered. The selectivity of top reviews, with their valences and parsimony, is therefore likely to influence customer satisfaction with the product. The signals from the valence of top reviews and those from all reviews are also expected to interact with each other and, depending on the level of their concordance, strengthen or weaken each other’s impact. As such, our framework analyzes the role that top review valences play in driving subsequent customers’ assessment of the information they gather and eventually their (dis)satisfaction with their purchase. Relatedly, we also study whether this impact is uniform across product popularity. With less popular products typically catering to a more heterogeneous customer base, it remains to be seen how top reviews impact the information efficiency in such cases.

Thus, in this paper, we address four related research questions: (1) What impact does the valence of top reviews have on improving the efficiency of information provisioning,

particularly when informational clutter increases? (2) What role does the parsimony in the number of featured top reviews play? (3) How does the concordance of information signals from *top reviews* and *all reviews* reinforce the persuasiveness of the signal of top reviews? (4) Is this impact uniform irrespective of the popularity of the product?

Analyzing a panel dataset collected from Amazon, we find that the impact of the valence of top reviews is amplified as the level of information overload increases, thus playing a role in clutter filtering. We then proceed to evaluate boundary conditions related to the quantity of information provisioned through top reviews. In particular, we leverage a quasi-natural field experiment on Amazon to document the diminishing return of top reviews when more of them are featured. In this setup, we compare a control group, in which a small but constant number of top reviews are featured, to a treatment group, where the number of top reviews changes over time, i.e., in which we vary the size of the set. We find that including three reviews in that set, which is the minimum level in our setting, outperforms any other larger review set. This leads us to conclude that an apparently minor change in the information display has major implications on information provisioning. For our assessment of the concordance of the signal of top reviews with all reviews, we devise a *signal reaffirmation* measure grounded in information theory that captures the concordance between the distribution of *top reviews* and *all reviews* valences. We find that such concordance plays a synergistic effect, supporting the signal from top reviews. As a robustness check, we supplement this analysis with a measure focused on reaffirmation in the content of top reviews and all reviews. Furthermore, we find that the role played by top reviews is differentiated across product popularity, with a weakened role for less popular products.

This study serves as a first step toward understanding the provisioning of information in online markets and the importance of top reviews and their signal reaffirmation. As such, we contribute to the literature on the nontrivial task of processing simultaneous signals in a complex informational environment. To the best of our knowledge, beyond acknowledging saliency (Archak et al., 2011), no prior study has focused on the role of top reviews in filtering out clutter and the role of their parsimony in determining the efficiency of information provisioning. As for concordance with other salient signals, the only related contemporaneous paper is Choi et al. (2019), which focuses, however, on the interactive effects of a social signal and a marketer signal without measuring their level of concordance. As such, retailers and users remain unable to fully capitalize on the value of reviews. Our paper contributes to this literature by exploring the avenues that a platform can use to improve the efficiency of its information provisioning.

Background

Information Overload in Online Platforms

Online reviews are key for providing customers with cues on important product dimensions and prior customer experiences (Babić Rosario et al., 2016; Brynjolfsson et al., 2003; Chen et al., 2008). By so doing, they reduce uncertainties related to product quality (e.g., Ellison & Fudenberg, 1995). On the flip side, review information is in overabundance, sometimes imperfect (Pavlou et al., 2007) and noisy (Branco et al., 2016; Dai et al., 2018; Luca, 2016), and can subject customers to information overload. As Jones et al. (2004) and, long before them, Rogers and Agarwala-Rogers (1975) wrote, information overload is the state in which not all communicated inputs can be processed and utilized by the receiver, leading to information breakdown. Such overload not only refers to the amount of information but rather to the combination of its quantity, ambiguity, and variety that a receiver has to process (Eppler & Mengis, 2004). As such, the large amount of information acts like noise, preventing the receiver from efficiently processing it (Klapp, 1986).

The information gathered by customers helps them reduce the uncertainty about their decisions; however, overloading them leads to inferior decision-making and a reduction in efficiency (Malhotra, 1982; O'Reilly, 1980). This problem is further compounded when the informational signals received—from reviews in our case—conflict with each other. As a result, the benefits of one source of information may effectively be canceled or crowded out by the other (Davis et al., 2021). In other words, customers experience the uncertainty of choice when information sources are contradictory (Broniarczyk & Griffin, 2014), leading to difficulties in evaluating both product attributes and fit (Hong & Pavlou, 2014; Kim & Krishnan, 2015). In turn, the efficiency of the information provisioned decreases, making the task of identifying the best product alternative more complex (Keller & Staelin, 1987). Some customers may still be able to navigate this complexity and make a purchase that ends up satisfying their needs. Others, however, may make an ill-fitting purchase (Hong & Pavlou, 2014) and experience a significant deviation from their a priori expectations (De et al., 2013), which they will eventually write about in a review. This breakdown in information provisioning is reflected in the dispersion of subsequent opinions that customers post about their purchases.

Information Signaling

In response to information uncertainty and overload issues, online platforms have developed a variety of signals about the products they sell. For example, platforms present crowd-sourced reviews to better inform customers about any product of interest. Future customers use these signals to become better informed and to guide their purchases. The purpose is to mitigate the breakdown in information provisioning resulting from overload and to better signal product quality. We, therefore, tap into signaling theory to explain this phenomenon: a product's unobservable quality is signaled through observable information that the platform synthesizes from crowd-sourced content. According to the broader understanding of the "Spencian" signaling theory (Connelly et al., 2011; Spence, 1973),² the term "signal" describes any informational characteristic, such as an attribute, that conveys information and credibly indicates (a product's) underlying quality, thus playing a role in decision-making.

In its original conception, signaling theory focuses on isolated signals with the unspoken assumption that receivers process signals in isolation (Bergh et al., 2014). In high-noise environments, however, where multiple signals are bundled together, this assumption is expected to no longer hold (Drover et al., 2018). Yet, most research studies have resorted to simply acknowledging that customers process multiple simultaneous signals about a prospective purchase in a complex informational environment (Dai et al., 2018; Nagle & Riedl, 2015) without studying signal interaction. Thus, prior research does not inform us about the role of selective signals, like top reviews, in decision-making under conditions of information overload. Another important factor influencing the effectiveness of signaling pertains to the concordance of multiple signals. Such concordance refers to the agreement between multiple signals for the same signaler (Chen & Rai, 2015; Gao et al., 2010). Signals that conflict with each other may confuse the receiver, resulting in less effective communication, while signals that are concordant may help mitigate those efficiency issues and reinforce signal persuasiveness (Fischer & Reuber, 2007).

Using the lens of signaling in the presence of information overload, we analyze in this paper the efficiency gains in information provisioning from top reviews. Specifically, we focus on their selectivity, parsimony, concordance, and differentiated role, which we detail next.

² For a detailed review of the use of signaling theory in management literature, please see Connelly et al. (2011).

Theory and Hypotheses

With customers receiving a number of simultaneous signals, processing those signals can become taxing. Customers, therefore, increasingly rely on a selective and salient signal to help them decipher suitable information to help them make fitting purchases. This fit level, however, may be hampered by the amount of noise surrounding the selective signal. It may also be hampered if the selective signal itself becomes noisy. Additionally, customers may use this signal in concordance with other available signals that may or may not reaffirm it. Yet empirical research has not investigated issues of signal selectivity, parsimony, and concordance on customer satisfaction. In this paper, we fill this gap and integrate theoretical constructs from information overload and signaling theory to formulate four hypotheses which we motivate and introduce next.

The Valence of Top Reviews and Information Overload

Researchers have long been interested in the impact of information overload on product choice. Early on, Friedman (1977) showed how, after certain levels of availability, information stops being useful. The “wealth of information” has been shown to lead to the “poverty of attention” (Simon, 1982), resulting in information overload and, thus, a reduction in decision-making effectiveness (Dobrajska et al., 2015). To mitigate this issue, the consensus was to provide information receivers with a single composite piece of information (Friedman, 1977) that is salient (Bordalo et al., 2013). The rationale is that the attention of decision makers is drawn to the most salient attribute(s) that stand out from others. Such salient signals have been shown to exert a powerful effect on the way people respond to messages and improving customers’ decisions (Hass, 1981). For example, featuring aggregate customer endorsement (in the form of “likes”) in the context of a music blog aggregator (Dewan et al., 2017) and in online coupon promotion (Li & Wu, 2018) has been shown to influence choices. In a context similar to ours, Chen et al. (2008) showed that featured reviews are more influential than those that are not.

Recognizing the influence of featured reviews, most review hosting platforms started featuring a select set of reviews based on crowd endorsement, typically derived from helpful votes, which we label *top reviews*. These reviews are typically extreme in their valence, either very negative or very positive (Forman et al., 2008). When more are positive in valence, customer satisfaction is affirmed. As such, when their overall positive valence increases, we expect more customers to make purchases that turn out to be satisfying for them; subsequently,

they are likely to review these purchases positively. Thus, products with higher demand and broader levels of satisfaction naturally receive a higher volume of reviews. As products start receiving more reviews, any negative review that makes it to the top reviews would understandably lower the overall valence of top reviews. More importantly, this would send a conflicting signal about the underlying product because a product with high demand is generally perceived less negatively. We would expect this to result in the deterioration of the efficiency of information provisioning. We therefore hypothesize:

H1: *The overall positive valence of top reviews improves the efficiency of information provisioning for products with more reviews.*

Top Reviews and Parsimony

With the significance of top reviews identified as leading to an increased level of information overload, the follow-up question is to investigate whether top reviews themselves can result in overload. Prospective customers typically face costs associated with processing information, making them resort to simplifying heuristics for solving complex problems. One such heuristic is to rely on parsimonious information. In fact, it is the simplicity of information that makes it most effective, especially when presented succinctly (Dai et al., 2018; Jones et al., 2004; Luca, 2016). Such concise and simple messages are superior in persuasion to complicated ones because they are easier to encode and decode faithfully (Corman et al., 2007). Additionally, responsiveness to information typically depends not only on the informational content but also on the simplicity of identifying the information of interest (Chetty et al., 2009).

Along these lines, and from early on, researchers have been interested in finding how much information to provide. In fact, in his highly influential paper with over 30,000 citations, Miller (1956) proposed that humans can process up to seven chunks of information (plus or minus two) which form the boundary of information processing and absolute judgment. Although his work is considered fairly preliminary and specific to one-dimensional judgments, it still points to an interest in quantifying how much information can be provided without causing overload. This decision on how concise the signal presented to customers should be lies in the hands of the platform, as it will significantly shape customer learning (Feldman et al., 2019). In our context, we label this conciseness of top reviews as parsimony and hypothesize that:

H2: *The parsimony of top reviews leads to an increase in the efficiency of information provisioning.*

Top Reviews and Signal Concordance

A burgeoning stream of research has emphasized moving away from the traditional signaling approach that focuses on the processing of signals in isolation (Drover et al., 2018). The aim instead is to account for the processing of the multitude of signals that are presented simultaneously. Most of the work in this area focuses on the firm setting and studies how firms signal their quality through a portfolio of metrics (Plummer et al., 2016). Crowdfunding entities, for example, publish portfolios of signals such as referrals, videos, text, and demonstrations that interact to mutually strengthen each other's impact, with the impact proportional to the level of inter-agreement (Steigenberger & Wilhelm, 2018). Building on this finding, we expect customers to assess the concordance of signals that they receive for their decision-making. They therefore assess the level of agreement between multiple signals (which we label *signal reaffirmation*). The intuition is that concordant signals are more reliable and, in turn, more conducive to decision-making (Boulding & Kirmani, 1993). Gao et al. (2010) label this agreement *signal consistency* in the context of software quality signaling. Chen and Rai (2015) also stress the importance of such signal consistency in the context of physicians' online popularity. The assumption is that conflicting signals are less effective because they confuse the receiver, while concordant ones reinforce the persuasiveness of signals (Fischer & Reuber, 2007). Furthermore, information integration theory stipulates that multiple information sources are more useful when they provide corroborating information than when they offer disparate conclusions (Anderson, 1981). On the other hand, when signals are in disagreement, confidence in judgments deteriorates (Lynch & Ofir, 1989) leading to misfit in choices and thus higher dispersion of subsequent opinions.

In the context of online purchases, platforms make available the full distribution of product reviews for each level on the rating scale (1 to 5 on Amazon). The reviews are also available to the customers. Additionally, customers are able to readily derive the distribution of the ratings of top reviews, given their limited set size (no more than 10 on Amazon). They, thus, are able to judge the level of agreement between the two distributions and associated signals. One would therefore expect that when the valences of top reviews and all reviews have distributions that mirror each other, customers are reassured by this concordance. The greater the concordance, the greater the reduction in uncertainty about product quality. We therefore hypothesize that:

H3: *The concordance of the signal of top reviews with that of all reviews leads to an increase in the efficiency of information provisioning.*

Top Reviews and Product Heterogeneity

The long-tail phenomenon in online channels has been well documented, showing that the distribution of online sales is less concentrated than in traditional channels, primarily because of the higher demand for less popular (niche) products (Brynjolfsson et al., 2009). Relatedly, product popularity moderates consumer decision-making and how reviews are consumed for different products (Duan et al., 2009; Zhu & Zhang, 2010). Generally, the popularity of products is measured based on sales (Brynjolfsson et al., 2009; Chen et al., 2021) and the demand for less popular products is more idiosyncratic. More importantly, less popular products entail greater uncertainty about their underlying features and, consequently, a larger variance in their reviews (Sun, 2012). Naturally, this higher uncertainty is expected to be reflected in top reviews as well, and their concordance with all reviews is also likely to be more uncertain. Consequently, even when the representative review signals (be it the valence of top reviews or their concordance with all reviews) are numerically similar, the uncertainty and variance is much higher for less popular products. It is well documented that higher uncertainty reduces the impact of the associated information signal (Kim & Krishnan, 2015; Pavlou et al., 2007). This suggests that top reviews would not be as effective for less popular products. In fact, Gu et al. (2013) show that positive reviews improve the sales of popular products more than that of less popular products. We, therefore, hypothesize that:

H4: *The signal from top reviews is less effective in improving the efficiency of information provisioning for less popular products.*

Research Design

Customers shopping at online retailers, like Target, Best Buy, and Amazon, receive more than one *signal* from reviews. Besides the signal from *all reviews* (including an average rating, as well as a volume and a histogram, see Figure 2), the other selective signal is from *top reviews* (displaying a selected set of reviews, see Figure 1). The platform enables customers to evaluate the related signals individually, which simultaneously helps to reduce their uncertainties about the product and make fitting decisions. In this context, we next develop measures and conduct analyses that capture the selectivity, parsimony, concordance, and differentiated role of top reviews in shaping the efficiency of information provisioning. We detail these measures next.

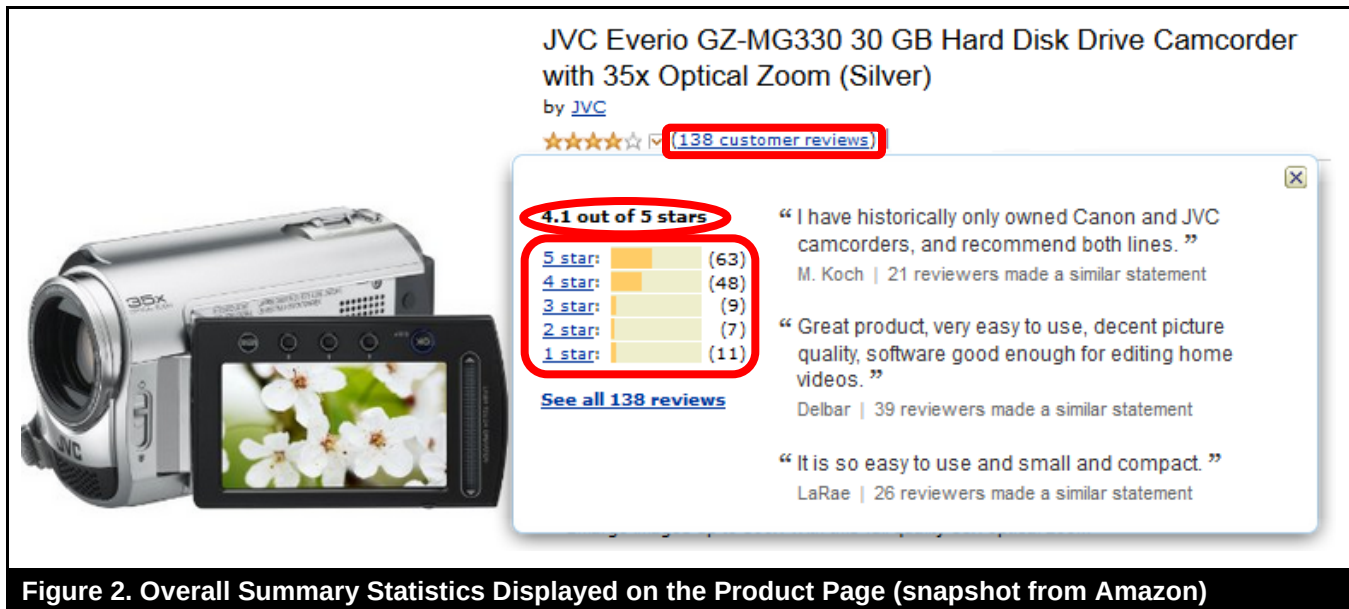


Figure 2. Overall Summary Statistics Displayed on the Product Page (snapshot from Amazon)

Main Measures

Measuring the Efficiency of Information Provisioning

One critical challenge that review hosting platforms face in their provisioning of information is customers' reduced discriminant ability to make fitting choices, which can sometimes lead to misfit in purchases. This is eventually manifested in the increased dispersion of subsequent opinions (in our context, in the form of a larger variance of subsequent ratings).³ We use variance to quantify this dispersion, as it has been frequently used in a variety of disciplines. For instance, Cotteleer and Bendoly (2006) use variance reduction in executing order specification and configuration to measure the effectiveness of an enterprise system implementation. Jia et al. (2020) use the variance of firm value to measure the effectiveness of an IS system automation implementation. Similarly, in financial economics, Jiang et al. (2005) and Zhang (2006) define information uncertainty regarding firm value in terms of subsequent variance in analysts' forecasts (which they label "value ambiguity"). As such, we label our main measure *dispersion of subsequent ratings* and operationalize it as the variance of new incoming ratings arriving at each time interval.⁴

³ We note here that a downstream spillover effect of the reduction in discriminant ability is reduction in demand (i.e., lower sales rank) which we discuss later in the paper. We thank the review team for pointing this out.

Measuring the Signal from Top Reviews

We propose a novel approach to measure the signal from top reviews. At any given time t , a product i has R_{it} reviews posted on its page with each such reviews receiving helpfulness votes. Given the difficulty of evaluating and synthesizing information from each of those R_{it} reviews and their accompanying helpfulness votes, it is boundedly rational to limit information gathering to a subset of top reviews $r_{it} \subseteq R_{it}$. The signal derived from this subset warrants further explanation.

Let $g(\bullet)$ be the mapping function that a customer employs to determine the signal of the product after reading the r_{it} top reviews. Naturally, not all r_{it} reviews have the same weight. The highest review in the r_{it} set, namely r_{it}^1 , with the most helpfulness votes from the crowd arguably has a more dominant influence on a customer than the reviews further down in the set. In other words, the influence of a review depends on its ranked sequence among top reviews. We capture the influence of a review at the η^{th} rank by its weight w_η (derivation detailed next). Then, to derive the weighted signal from the r_{it} top reviews, we assume a linear mapping of review ratings, which we weight by w_η . This mapping function is assumed to be uniform across customers and is defined as follows (Equation 1):

⁴ To ensure the validity of our measure, we restricted the products in our study to those that were active in terms of sustained arrival of new incoming reviews. Specifically, we set a minimum threshold of at least five new reviews arriving in each time interval.

$$g(r_{it}) = \frac{\sum_{\eta=1}^{|r_{it}|} w_{\eta} V_{\eta}}{\sum_{\eta=1}^{|r_{it}|} w_{\eta}}, \quad (1)$$

where V_{η} is the rating of the η th rank-ordered top review, w_{η} is the weight of the η th rank-ordered top review, and $|r_{it}|$ is the overall number of top reviews.

The derivation of review weights w_{η} merits further explanation. As stated earlier, several online sellers enable the crowd to endorse a review by casting a vote of helpfulness. The retailer then features a series of those top reviews listed in a sorted order based on those votes. In our aggregation approach, the weight w_{η} of review η is set as its reverse rank order (*Reverse Rank Order* _{it}) in the subset r_{it} , weighted by the total number of top reviews $|r_{it}|$. It is therefore calculated as $w_{\eta} = \text{Reverse Rank Order}_{it} / |r_{it}|$, making the weight monotonically smaller with a smaller reverse rank order.

The intuition behind this approach is that, rather than weighing a review directly with the actual but complex helpfulness vote listed next to each review (e.g., a given review showing that “321 out of 335 people found the following review helpful” which we believe is complex to assimilate), we use a rather simple but intuitive proxy, the reverse rank order of the review out of the total number of reviews received (e.g., the top review from among 10 reviews receives a rank order of 1, or a reverse rank order of 10, leading to a weight of 10/10, and the second top most review among those 10 reviews receives a rank order of 2, or a reverse rank order of 9, leading to a weight of 9/10). Derived in this way, w_{η} accounts for two forces. First, the use of the reverse rank order ensures that reviews that have floated to the top are more heavily weighted. Furthermore, this rank weighs the reviews that floated to the top among a large pool of reviews (say ten) more heavily than those that floated to the top among a smaller pool (say three).

Measuring Signal Reaffirmation

This metric measures the concordance between the valences of top reviews and those of all reviews. As such, it captures the stochastic distribution underlying the review ratings. We use *mutual information* (MI), a well-established information-theoretic measure (Shannon, 1948) that accounts for the reduction in the uncertainty of one random variable due to knowledge of the other random variable (MacKay, 2003). MI is a nonnegative measure that is equal to zero when the two

random variables studied are statistically independent. It has been recognized as a global measure of association that incorporates more information than traditional linear measures of correlation (Granger & Lin, 1994) to quantify the strength of dependence between two variables (Granger et al., 2004; MacKay, 2003). In information systems, it has been used to identify informative attributes for predicting bank failure (Sarkar & Sriram, 2001), to measure the distance between opinions in reviews (Archak et al., 2011), and to assess software diversity (Temizkan et al., 2017). It has also been used extensively in other disciplines, including finance, to measure serial dependence in stock markets (Ferreira & Dionisio, 2014), and in economics to quantify the realized value of acquiring a signal (Boyaci & Akcay, 2018). We adopt MI to quantify the extent of concordance between top reviews and all reviews and label it *signal reaffirmation*, thus indicating the strength of the signal from top reviews.⁵

One may be tempted to think about this reaffirmation between two signals in terms of the correlation between top reviews and all reviews. This, however, is adequate from neither a relevance dimension nor an implementation dimension (see additional explanation about the shortcomings of correlation in Taleb, (2019)). From a relevance dimension, correlation is restrictive in that it measures the *linear* dependence between two variables. As such, an increase in value of, for example, 0.1 when the correlation is, say, 0.3 is interpreted the same way as when the correlation is, say, 0.9 (when that is obviously not the case). MI, on the contrary, does not make that restrictive assumption. From an implementation dimension, the correlation between two variables, say, X (for top reviews) and Y (for all reviews) requires an identical number of instances for both variables X and Y and requires that those instances are pairwise matched to derive their pairwise covariances and then their standard deviations. Of course, this is not possible in our context, as there are almost always significantly more reviews in the *all reviews* category than in the *top reviews* category. MI does not impose those restrictions.

Next, we outline the metric derivation. There are 5 events corresponding to observing 1, 2, 3, 4, and 5-star reviews, and there is a probability attached to each of those five events. For *all reviews*, these probabilities can be readily derived from data using the ratio of the number of stars at this rating (i.e., 1, 2, 3, 4, or 5) and the total number of reviews. Additionally, these probabilities are intuitively available to customers through the histogram that online sellers display. These probabilities attached to 1, 2, 3, 4, and 5-star reviews can also be readily derived for top reviews, as the corresponding

⁵ For the derivation of MI, we did not require customers to consult the rating of each of the reviews in all reviews. With our measure based just on the distribution of all reviews, we note that this distribution is readily available

to customers. On Amazon, for example, the rating histogram is readily displayed when customers hover over the product star rating.

number of reviews is much smaller and thus can be derived with ease. MI varies between 0 and infinity. As such, MI is a relative rather than absolute measure: the higher the value of MI, the more related the variables are, thus the more reaffirmation there is.⁶

Computationally, deriving the *signal reaffirmation* requires the joint distribution of ratings of *top reviews* as well as that of *all reviews*. As those joint distributions are not readily available, we rely on the marginals for imputation, making use of methodological development in statistical theory and practice. These disciplines have provided an active area of research for close to a century for imputing a joint distribution of variables based on their marginal distribution with wide-ranging implementations in physics, economics, engineering, risk analysis, and medicine (Cicalese et al., 2017; Gomez-Deniz et al., 2011; Lin et al., 2014). Because of inherent computational challenges, however, only a handful of nonrestrictive methods have been successfully proposed for the imputation. In this paper, we follow the recent advancements made by Frogner and Poggio (2019),⁷ who define a general-purpose inference algorithm that is computationally efficient, applicable to real data, and superior in performance to existing approaches. We implement their approach (see Algorithms 1 to 5 in their paper) to derive the joint distribution, and then use this derivation to calculate the MI between top reviews and all reviews for each product at time *t*, which we label *signal reaffirmation*. We provide additional details in Appendix A.

Alternative Measure of Signal Reaffirmation

It may be the case that review content also plays a role in signal reaffirmation. For instance, customers may read the content of top reviews and figure out the main topics considered in those reviews, then read the content of all reviews to see how representative the topics of the top reviews are compared to all reviews. As such, the level of concordance (or lack thereof) between the content of those two sets of reviews may also be used as a measure of signal concordance. As a robustness check, we therefore derive an alternative measure of signal reaffirmation between top reviews and all reviews but this time based on the level of agreement in the textual content of these reviews. Specifically, this measure is calculated as the percentage of topics mentioned in *top reviews* out of all the topics mentioned in *all reviews*. The derivations are provided in Appendix B. However, this measure may be quite noisy. For one thing, customers are not expected to read every single review in the *all reviews* set.

⁶ We note that the MI median is 0.54 in our data. As this variable is unbounded, such a low median value is indicative of a large variation between Top and All Reviews. Additionally, 18% of the observations are equal to zero, also indicative of large variation.

Additionally, the interpretation of the topics considered in the reviews may vary from one customer to another. That said, this alternative measure will still shed some light on the influence of concordance on information provisioning.

Additional Variables

Following the related literature on word-of-mouth marketing (Babić Rosario et al., 2016), we also include a number of control variables that are proxies for other contemporaneous factors: the mean of all reviews (*all reviews mean*), widely acknowledged as an overall quality signal that platforms prominently display to customers, total count of reviews received (*all reviews count*), signal consistency between top reviews (*signal consistency*), which is derived as the entropy of the information in the text of the top reviews (derivation provided in Appendix B), standard deviation of ratings (*all reviews std. deviation*), duration the product has been listed on the platform (*days since available*), product sales rank (*sales rank*), sale price (*price*), ratio of the count of one-star reviews to the count of five-star reviews (*ratio rating extremes*), total number of images posted in reviews (*images posted*), total number of videos posted in reviews (*videos posted*), a dummy for whether the product is sold directly by the platform (in our case, Amazon) (*Amazon seller*), a dummy for whether the product is shipped by the platform (*Amazon shipper*),⁸ and a dummy for the product categories. We note that *days since available* may also be considered a proxy for the newness of the product and thus its appeal to customers. *Ratio rating extremes* accounts for review disagreement and is in line with findings that emphasize the role of disagreement in extreme review ratings over time (Nagle & Riedl, 2015). Using a ratio captures the mix of these two extreme ratings. Table 1 provides a description and summary statistics of our variables, and Appendix C presents the correlation matrix.

Data Description

Our data sample was obtained from Amazon, the largest U.S. online retailer and a leader in online review systems. We focused on four categories, namely Automotive, Grocery/Gourmet Food, Health/Personal Care, and Home/Kitchen. We then identified ten subcategories; one such subcategory for example was Cooking & Baking under the category Grocery & Gourmet Food.

⁷ It is worth noting that the author Tomaso Poggio has received more than 116 thousand citations of his papers.

⁸ We note that, as more reviews are posted, the number of images and videos added changes over time. Similarly, products do change sellers and shippers from time to time.

Table 1. Variable Definition and Descriptive Statistics

Variable	Description	Mean	Std. dev.	Min	Max
<i>Dispersion of subsequent ratings</i>	Ratings variance of subsequent reviews	1.00	0.88	0	4
<i>Top reviews mean</i>	Ratings mean of <i>top reviews</i>	4.14	0.85	1	5
<i>Signal reaffirmation</i>	Agreement between top reviews and all reviews and measured as their mutual information	0.51	0.39	0	1.91
<i>Text-based signal reaffirmation</i>	Percentage overlap between topics mentioned in top reviews and those mentioned in all reviews	0.54	0.22	0.3	1
<i>All reviews mean</i>	Ratings mean of all reviews	4.37	0.33	1.2	5
<i>All reviews count</i>	Total number of reviews received (log transformed)	6.76	1.12	0.69	9.94
<i>Signal consistency</i>	Entropy of the textual content of top reviews based on topics and sentiments (multiplied by -1)	3.04	0.49	0	3.95
<i>All reviews std. deviation</i>	Standard deviation of review ratings	1.08	0.24	0	1.78
<i>Days since available</i>	Total number of days product listed (log transformed)	7.24	0.73	4.34	8.65
<i>Sales rank</i>	Sales rank (log transformed)	5.42	2.19	0	14.11
<i>Price</i>	Product price (log transformed)	2.74	1.11	0	6.22
<i>Ratio of rating extremes</i>	Ratio of the count of 1-star to 5-star reviews (log transformed)	-2.64	0.91	-5.88	3.23
<i>Images posted</i>	Total number of images posted (log transformed)	6.98	7.36	0	10
<i>Videos posted</i>	Total number of videos posted (log transformed)	4.62	5.31	0	8
<i>Amazon seller</i>	Dummy variable for products sold by Amazon (1 if Yes, 0 Otherwise)	0.37	0.48	0	1
<i>Amazon shipper</i>	Dummy variable for products shipped by Amazon (1 if Yes, 0 Otherwise)	0.71	0.45	0	1

Note: Dataset 1 has $N = 56,460$ observations with *Panels* = 919 products

Within each subcategory, an average of 85 products were selected for the final product sample set. We designed crawlers that visited the individual product pages and scraped the relevant content twice a week over a nine-month period between February and November 2015.⁹ This time window is sufficiently comprehensive, allowing us to track products that have been selling on Amazon anywhere from between a few weeks to a few years. We collected the product description, price, sales rank, mean rating, and reviews received, as well as the helpfulness vote of each review associated with the product. We also collected top reviews that the retailer selected to feature to customers. Thus, the panel dataset consisted of all the review-related information associated with the aforementioned products, as well as the progression of this information over time.

⁹ With the set of top reviews frequently changing over time, our data collection strategy consisted of visiting the retailer's product pages and collecting the live information about reviews before they are updated. Accordingly, we designed our data collection in a way that allowed revisiting the site for updated content twice a week to capture the changes in the sequence of top reviews featured on the product page and the product sales rank and at each data collection instance.

To ensure data consistency, we dropped duplicate product listings.¹⁰ We also dropped all products with missing data such as sales rank or price. Additionally, we ensured that all the products had a sales rank reported at the category level to ensure a valid comparison across products. Lastly, we only considered products with five or more reviews in each time interval. This is a reasonable restriction that ensures an adequate level of variability in the derivation of our variables. This dataset is further detailed in Table 1.

To investigate H2, and to further validate our main findings, we used another dataset from Amazon, which covers a relatively shorter time span (Dataset 2). It consists of electronics categorized as *cameras*, *camcorders*, and *tablets* in the department of Cameras & Photo and Computer & Accessories.¹¹ These products are known to have dynamic

¹⁰ The product catalog includes multiple listings with unique SKUs that are minor variations of one another. One such example is the variety of colors for a given camera model. The retailer maintains a unique SKU for each product version but pools the reviews for all the related versions. We therefore ensured that we removed all duplicates and kept only one product version in the dataset.

¹¹ The collection for this dataset was performed between July and October 2013.

market performance and receive a large number of reviews. This dataset is further described in Table 3. We note here that while collecting the data, we coincidentally observed a change in the display of top reviews, as Amazon had adjusted the number of top reviews it featured on the product page for a subset of products. We used this feature subsequently in the quasi-natural experimental setup we analyze in the Signal Efficiency and Parsimony section below.

Empirical Models and Results

Our analysis investigates the effect of information overloading in the presence of top reviews on the efficiency of information provisioning, which we operationalize as *dispersion of subsequent ratings*. We included two key signals: the selective signal from top reviews (*top reviews mean*) and the concordance between top reviews and all reviews (*signal reaffirmation*). Our estimation accounts for a variety of proxies, as detailed earlier. We also included a number of observed and unobserved factors, detailed next, that could influence the dynamics of review posting and determine why a given product receives certain review ratings at a given time period. Accounting for these dynamics can help to overcome biases in the estimate of main effects. Furthermore, some of these signal measures are dynamic and may be correlated with their past realizations. To overcome these identification challenges, we used the Arellano-Bover/Blundell-Bond (AB/BB) (Arellano & Bover, 1995; Blundell & Bond, 1998) dynamic panel model estimation (see, for example, Roodman, 2006; Bapna et al., 2013).^{12,13} This dynamic panel model is a single equation model with autoregressive dynamics and explanatory variables that are not assumed to be strictly exogenous. This model has been extensively used in IS and other economics subfields where a generalized method of moments (GMM) estimator is used to consider all of the potential orthogonality conditions for addressing cases like ours (Bhargava & Mishra, 2014; Menon & Kohli, 2013). Our model is expressed in Equation (2):

$$\begin{aligned} \text{Dispersion of subsequent ratings}_{i,t+1} = & \beta_1 \text{Top reviews mean}_{it} + \mu_1 \text{Signal reaffirmation}_{it} + \\ & \varphi_1 \text{All reviews mean}_{it} + \pi_1 \text{All reviews count}_{it} + \\ & \theta_1 \text{Dispersion of subsequent ratings}_{it} + \delta_{c1} \text{Control} + \\ & \alpha_i + \varepsilon_{it}, \end{aligned}$$

where i refers to the product and t refers to the biweekly observation, *Dispersion of subsequent ratings* _{$i,t+1$} is the variance of review ratings between t and $t+1$, *Top reviews mean* _{it} is the weighted mean of top reviews ratings, *Signal reaffirmation* _{it} is the mutual information between top reviews and all reviews ratings, *All reviews mean* _{it} is the mean of all reviews ratings, *All Reviews Count* _{it} is the total count of all reviews, *Control* is a vector of c control variables, α_i is a product-specific effect, and ε_{1it} is the error term. Control variables consist of *all reviews standard deviation*, *days since available*, *sales rank*, *price*, *ratio of rating extremes*, *images posted*, *videos posted*, *Amazon seller*, and *Amazon shipper* (as detailed in Table 1). The correlation matrix is in Appendix C.

For our estimation, we used the fixed-effects Arellano-Bond general method of moments' (AB-GMM) dynamic implementation augmented by (Blundell & Bond, 1998) and (Windmeijer, 2005) and adopted a two-step robust estimation. Specifically, we implemented the system GMM estimation which uses a system of two equations, one with the level variables and another transformed by first differencing to identify parameter estimates. We conduct the second-order Arellano and Bond (1991) test for serial correlation in differences. The test entails examining whether the differenced error term is second-order and serially correlated with the AR(1) expected in first differences. Whenever those conditions were met, we did not reject the null of no second-order autocorrelation in the first-differenced errors. We also tested for heteroskedasticity and applied the Windmeijer correction to correct for the bias in two-step standard errors estimating robust errors. We also tested the instruments' exogeneity assumption using the Hansen (1982) J test, a robust variant of the Sargan test of overidentifying restrictions. Regarding the number of instruments to use, Roodman (2009) proposed a rule of thumb restricting the number of instruments to be fewer than the number of units to avoid downward-biasing of the standard errors. In our setup using a large sample size, we had adequate flexibility in selecting the number of instruments to include (Roodman, 2006). We, therefore, used up to ten lags of the dependent variable as instruments for the transformed equation wherever appropriate and used contemporaneous first differences as instruments in the level's equation. To test our hypotheses, we provide different variations of Equation (2) and include the related statistics and results in subsequent tables.

¹² Arellano-Bond (AB) estimation starts by transforming all regressors by differencing using *generalized method of moments* (GMM), or difference GMM. The Arellano-Bover/Blundell-Bond estimator augments the AB estimator by making an additional assumption that first differences of instrument variables are uncorrelated with the fixed effects. This allows for the introduction of more instruments and improves efficiency. As such, it

builds a system of two equations—the original equation and the transformed one which transforms the instruments by differencing to make them exogenous to the fixed effects.

¹³ The term "dynamic" refers to adding the lagged dependent variable as a regressor in the equation (Baltagi, 2008).

Signal Efficiency and Selectivity

We start by investigating the effect of information overload resulting from the increasingly large review volume posted on product pages and the corresponding implications of the valence and selectivity of top reviews. To that end, we modify Equation 2 and include an interaction term between the variables *top reviews mean* and *all reviews count* (i.e., *Top reviews mean* $\times$ *All reviews count*):

$$\begin{aligned} \text{Dispersion of subsequent ratings}_{i,t+1} = & \beta_1 \text{Top reviews mean}_{it} + \tau_1 \text{Top reviews mean}_{it} * \\ & \text{All reviews count}_{it} + \mu_1 \text{Signal reaffirmation}_{it} + \\ & \varphi_1 \text{All reviews mean}_{it} + \pi_1 \text{All reviews count}_{it} + \\ & \delta_{c1} \text{Control} + \theta_1 \text{Dispersion of subsequent ratings}_{it} + \\ & \alpha_i + \varepsilon_{it} \end{aligned}$$

In Table 2 we present the results of our econometric estimation to explore support for H1. The coefficient on *top reviews mean* in Column (2.1) is negative and significant ($\beta_1 = -0.591$ at $p < 0.05$). This indicates that, as the valence of top reviews becomes increasingly positive, the dispersion in subsequent ratings decreases, signifying a better efficiency in information provisioning. Equally importantly, the coefficient associated with the interaction term *Top reviews mean* $\times$ *All reviews count* is significant and negative ($\tau_1 = -0.116$ at $p < 0.05$), indicating that at higher review volumes, i.e., when there is a higher potential for information overload from reviews, the impact of *top reviews mean* is even stronger. Thus, we find support for H1. The combined effect from the variable and its interaction amounts to a decline of 0.25 in review dispersion when the values of *top reviews mean* and *all reviews count* increase from their 50th to 75th percentile values.

We supplement our analysis of the role of the selectivity of top reviews in mitigating the information overload associated with an increasing number of all reviews by reestimating the same model specification, this time adopting a subsampling approach. We formulate the first subsample to get a rather small number of all reviews per product, with each of the products receiving no more than 10 all reviews in total. The second subsample consists of products that received relatively more all reviews per product, with each of the products receiving fewer than 25 all reviews in total. The third subsample consists of products that received substantially more all reviews per product, with each of the products receiving more than 100 all reviews in total. The results across all columns are qualitatively similar, providing further support for H1. For instance, inspecting Column (2.4) with products receiving more than 100 reviews, the key variable of interest, *top reviews mean*, is significant and negative ($\beta_1 = -0.508$ at $p < 0.01$), indicating that as the signal from top reviews increases, the dispersion of subsequent ratings decreases. This amounts to a 0.22 decline in

the review dispersion when the value of top reviews increases from its 50th to its 75th percentile value. It is also noticeable that β_1 becomes increasingly larger as we move from Columns (2.2) to (2.3) to (2.4), and is always significant and negative, pointing to the larger role that top reviews play as information availability (i.e., more all reviews) and thus overload increases. It is worth noting that the coefficient on *all reviews mean* is significant and negative for all the regression specifications, a finding consistent with existing studies (Archak et al., 2011; Chevalier & Mayzlin, 2006).

Signal Efficiency and Parsimony

The natural follow-up question pertains to whether the effectiveness of the signal of top reviews may itself dwindle if too many such reviews are featured, thus overwhelming customers. The concern, of course, is that too many such top reviews would potentially result in information overload, eventually hampering their capacity to filter out clutter. In other words, the question becomes whether the value from top reviews is diminished when the number of such reviews selected is no longer parsimonious. To assess this concern, we investigated the effectiveness of top reviews when the platform chooses between the display of a limited number of top reviews and the display of a larger number of such reviews. The importance of this type of analysis is that it allows us to provide guidelines on how signal effectiveness is impacted when the level of information featured is variable and on the impact of ensuring parsimony in the set of top reviews.

To answer this question, we were fortunate to leverage a quasi-natural field experiment related to top reviews selectivity. As noted before, while we were collecting a second dataset (Dataset 2), Amazon adjusted the number of top reviews it featured on the product page for a subset of its products—this display adjustment did not occur in Dataset 1. As such, among the products we tracked, some maintained a constant and small number of top reviews (three such reviews) while others started at that same constant level of three for a period of time before the platform changed that set size to fluctuate over time at levels larger than three. Note that the minimum level of three was set by Amazon. This setup makes it amenable to a quasi-natural field experiment that investigates the impact of a top reviews set size above a base level. As such, the control group included all products in our dataset that maintained the small and constant number of three top reviews during the study period. The treated group included all products in our dataset that had a set size for top reviews that was larger than the control group during the post-treatment period. Consequently, we utilized the difference-in-differences (DID) approach, combined with matching, to isolate the effects.

Table 2. Dispersion of Subsequent Ratings Regression Results: Review Volume

DV: Dispersion of subsequent ratings	All Products (2.1)	Products with <10 reviews (2.2)	Products with <25 reviews (2.3)	Products with >100 reviews (2.4)
Top reviews mean	-0.591** (0.309)	-0.386** (0.203)	-0.428*** (0.125)	-0.508*** (0.221)
Top reviews mean × All reviews count	-0.116** (0.06)	-	-	-
Signal reaffirmation	-1.395*** (0.529)	-0.516* (0.274)	-0.711* (0.357)	-0.926** (0.406)
All reviews mean	-0.547** (0.3)	-0.91*** (0.273)	-0.56** (0.29)	-0.696*** (0.23)
All reviews count	-1.853* (1.125)	-2.363** (1.068)	-1.577*** (0.612)	-1.962*** (0.568)
Signal consistency	-0.464 (0.584)	-0.438 (0.438)	-0.487** (0.211)	-0.403** (0.203)
All reviews std. deviation	-0.338* (0.203)	-0.411** (0.191)	-0.138 (0.282)	-0.233* (0.121)
Days since available	-0.05 (0.105)	-0.07* (0.038)	-0.058 (0.118)	-0.037 (0.025)
Sales rank	-0.09* (0.053)	-0.095*** (0.01)	-0.087*** (0.028)	-0.044* (0.025)
Price	0.244 (0.183)	0.187* (0.102)	0.051 (0.18)	0.076 (0.219)
Ratio of rating extremes	0.306 (1.515)	0.634 (3.016)	1.283 (2.897)	2.222 (2.286)
Images posted	0.91 (3.077)	1.388 (1.547)	1.133 (1.71)	1.238* (0.681)
Videos posted	-1.184 (2.673)	-0.457 (1.541)	-0.64 (2.996)	-0.581 (0.962)
Amazon seller	-0.038 (0.161)	-0.027 (0.06)	-0.013 (0.039)	-0.038 (0.076)
Amazon shipper	0.112 (0.228)	0.102 (0.29)	0.118 (0.199)	0.11 (0.323)
DV lag	Y	Y	Y	Y
Product-fixed effects	Y	Y	Y	Y
Model fit statistics				
Wald χ^2	***	***	***	***
Prob. auto-regressive				
AR(1)	***	***	***	***
AR(2)	0.41	0.70	0.65	0.71
Hansen test of overidentification				
P	0.78	0.88	0.64	0.87
Panel	919	193	453	114
Observations	56,460	11,857	27,831	7,004

Note: Variables are derived for each product i and time t . Independent variables are lagged. DV lag is a lag of the dependent variable. Estimation is based on Equation (2) using AB/BB GMM. Column (2.1) shows the results for all products; Column (2.2) for products receiving less than 10 reviews per product; Column (2.3) for products receiving less than 25 reviews per product; Column (2.4) for products receiving more than 100 reviews per product. Regression includes 73 instruments. Standard errors are listed in parentheses. Significance is *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Sample

In this sample, we identify the full assortment of products in three categories: *cameras*, *camcorders*, and *tablets*, consisting of products listed on the site anywhere between a few weeks to a few years. We tracked these products over a period of 16 weeks between July and October 2013 and conducted an exhaustive data collection gathering (data elements identical to those in Dataset 1). The resulting panel dataset consists of all the review-related information associated with the aforementioned products, as well as the progression of this information over the 16 weeks. We provide all the relevant summary statistics in Table 3.¹⁴

Model

To estimate the effect of the parsimony of the set of top reviews, we contrast our outcome variable, *dispersion of subsequent ratings*, for products with a parsimonious and constant number of top reviews (control) and those products with a large and variable number of top reviews (treatment). Our DID strategy, therefore, identifies the treatment effect of increasing the number of top reviews. In our specification, we included a set of controls identical to those used in the main analysis. We adopted an intensity-based DID design (Hirano & Imbens, 2004), i.e., we utilized the number of top reviews featured for each product at each time period. Accordingly, the treatment variable is categorical, *TreatmentCateg*, reflecting the count of top reviews. The complete model specification for the treatment effect on *dispersion of subsequent ratings* is a modification of Equation (2):

$$\begin{aligned} \text{Dispersion of subsequent ratings}_{i,t+1} = & \theta_{1,did} \text{TreatmentCateg}_{it} + \beta_{1,did} \text{Top reviews mean}_{it} + \\ & \mu_{1,did} \text{Signal reaffirmation}_{it} + \\ & \gamma_{1,did} \text{Signal reaffirmation}_{it} * \text{All reviews mean}_{it} + \\ & \phi_{1,did} \text{All reviews mean}_{it} + \delta_{c1,did} \text{Control} + \alpha_i + \varepsilon_{1,did,it} \end{aligned}$$

Matching and Findings

A concern related to any DID analysis may be that we do not actually assign the products to the control and treatment groups and therefore cannot rule out nonrandom assignments. To mitigate this concern, we utilized matching on observable characteristics. The objective of matching is to identify products in treatment and control groups that resemble each other in all relevant characteristics before the change in the number of top reviews (the treatment). We aimed for statistical equivalence between the two groups by balancing them on observables.

We began by matching treated products to untreated ones to ensure that our treatment and control groups were comparable. To avoid contamination of our treatment, we only considered products that started at the same level as that of the control group for two time periods and then started fluctuating but never returned to the original control level. We then identified a set of variables, which we believe ensures balance across the treatment and control groups through bivariate tests. The variables selected for matching, along with the related statistics, are reported in Appendix D. We present the main estimates with matching based on the nearest neighbor algorithm.¹⁵

The regression coefficient on the variable of interest *TreatmentCateg* in Table 4 confirms our hypothesis. This coefficient is significant and positive ($\theta_{1,did} = 0.028$ at $p < 0.05$) indicating that switching from a constant small number of top reviews to a larger number (more than three) results in an increase in the dispersion of subsequent ratings. Our findings provide support for H2. They validate the need for information and its signals to be presented simply and succinctly to make them more effective. Otherwise, prospective customers face the cost of processing additional information and that additional cost hurts rather than benefits the efficiency of information provisioning.

DID Validation

While we have no direct statement from Amazon, or clear indication that this quasi-experiment is indeed being implemented, we were still able to conduct a battery of checks to ensure the validity of the difference-in-differences approach and the exogeneity of the shock. We conducted model-free raw data investigations of the control and treatment groups to ensure that these two groups were similar before treating them on a broad range of attributes (including sales rank). For our dependent variable *dispersion of subsequent ratings*, we graphically checked the parallel trend prior to the start of the experimental setup. The trends displayed are quite similar before treatment started, giving us confidence in the DID design. We provide those details in Appendix D. We believe this evidence supports the assumption that in the absence of treatment, the average outcomes for the treated and control groups would have followed parallel trends over time (Abadie, 2005). Additionally, to validate our quasi-experimental setup, we conducted matching between control and treatment units using alternative matching algorithms.

¹⁴ In addition to conducting our DID analysis, we repeated the main analyses with this alternative dataset to ensure that the findings are qualitatively similar. We report these findings in Appendix G.

¹⁵ We allow four nearest neighbors.

Table 3. Descriptive Statistics for Dataset 2

Variable	Mean	Std. dev.	Min	Max
<i>Dispersion of subsequent ratings</i>	0.60	0.82	0	3.61
<i>TreatmentCateg</i>	5.33	1.86	3	10
<i>Top reviews mean</i>	4.10	0.85	1	5
<i>Signal reaffirmation</i>	0.60	0.32	0	1.40
<i>All reviews mean</i>	4.05	0.46	2.55	5
<i>All reviews count</i>	5.52	1.13	1.79	8.08
<i>Signal consistency</i>	2.96	0.38	0	3.56
<i>All reviews standard deviation</i>	1.22	0.26	0	1.72
<i>Days since available</i>	5.96	0.66	3.56	8.54
<i>Sales rank</i>	4.21	2.25	0	11.04
<i>Price</i>	5.43	0.77	3.61	7.60
<i>Ratio of rating extremes</i>	-1.94	1.10	-4.48	0.69
<i>Images posted</i>	123.46	225.46	1	1,356
<i>Videos posted</i>	13.36	18.47	0	88
<i>Amazon seller</i>	0.31	0.46	0	1
<i>Amazon shipper</i>	0.63	0.48	0	1

Note: Dataset 2 has $N = 14,857$ observations with *Panels* = 596 products.

Table 4. Dispersion of Subsequent Ratings Regression Results DID Model with Categorical Treatment

DV: <i>Dispersion of Subsequent Ratings</i>	DID (4.1)
<i>Treatment (categorical)</i>	0.028** (0.012)
<i>Top reviews mean</i>	-0.63*** (0.147)
<i>Signal reaffirmation</i>	-0.132*** (0.048)
<i>All reviews mean</i>	-0.326** (0.145)
<i>All reviews count</i>	-0.087 (0.073)
<i>Signal consistency</i>	-0.421** (0.213)
<i>All reviews std. deviation</i>	-1.71 (1.359)
<i>Days since available</i>	-0.156 (0.095)
<i>Sales rank</i>	-0.035** (0.017)
<i>Price</i>	0.248** (0.12)
<i>Ratio of rating extremes</i>	0.289 (0.226)
<i>Images posted</i>	0.193 (0.121)
<i>Videos posted</i>	-0.013 (0.023)
<i>Amazon seller</i>	-0.013 (0.018)
<i>Amazon shipper</i>	0.032 (0.024)

Time-fixed effects	Y
Product-fixed effects	Y
Model fit statistics	
R^2	0.52
Panel	596
Observations	14,857

Note: Variables are derived for each product i and time t . Independent variables are lagged. The treatment is a categorical variable. Estimation uses robust OLS with a product- and time-fixed effects. Matching is based on the Nearest Neighbor. Standard errors are in parentheses. Significance is *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Because each matching algorithm is subject to some limitations, replicating the analysis with a variety of algorithms boosts confidence in our findings. We provide these regression results in Appendix E. In Appendix F, we also include our findings from a random treatment implementation analysis that shuffles the treatment indicators in the dataset to a randomly selected set of control observations, thereby creating a placebo treatment that ensures that the results are not due to spurious relationships. The results give us confidence that the treatment effect is unlikely to be spurious.

Signal Efficiency and Concordance

We now turn to the role of *top reviews* and their signal concordance, thus studying the implication of simultaneously presenting signals from top reviews along with all the other reviews (*all reviews*) on the efficiency of information provisioning. In Table 5 we present the results of our estimation of Equation (2) (Dataset 1 is used). We estimate two versions that differ in the way that *signal reaffirmation* is derived. In Column (5.1), we use the derivation from the Measuring Signal Reaffirmation section above based on the mutual information between the rating distribution of top reviews and all reviews. While the key variable *top reviews mean* remains significant and negative ($\beta_1 = -0.599$ at $p < 0.05$), the other key variable of interest, *signal reaffirmation*, is also significant and negative ($\mu_1 = -0.476$ at $p < 0.01$), supporting H3. This implies that the concordance between the signals from top reviews and all reviews is critical for the efficiency of information provisioning through reviews. Economically, we find that an increase of a quarter of a star in *top reviews mean*, combined with an increase of 0.1 in *signal reaffirmation*, leads to a reduction of the dispersion of subsequent ratings of 0.2.¹⁶

As for the control variables, the variables *all reviews mean* ($\phi_1 = -0.649$ at $p < 0.01$) and *all reviews count* (-1.43 at

$p < 0.1$) are negative and significant, a hardly surprising finding, considering their well-documented influential role. *Signal consistency* (-0.48 at $p < 0.01$) is also significant, an indication that, on average, the content of top reviews and their consistency is also influential along with the other aggregated signals. Furthermore, the coefficient for *all reviews std. deviation* is also significant and negative (-0.287 at $p < 0.1$). The coefficient on the *signal ratio of rating extremes*, which is a good measure of extreme polarity in ratings is, however, not significant. (Table 5).

Reassuringly, the results are qualitatively similar. In fact, the magnitude is larger for the coefficient on *signal reaffirmation*. This reemphasizes the importance of concordance between the top reviews and the information content in all reviews.

Signal Efficiency and Product Popularity

To address the question of a differentiated role of review signals based on product popularity, we classify products based on their sales levels. To that end, we label products as *more popular* if their sales rank is in the top 100. All the remaining products are labeled as *less popular*. For this labeling, we rely on the product sales rank observed at the beginning of our data collection.¹⁷ The summary statistics for both groups are presented in Table 6. We reestimate Equation (2) separately for each of the popularity subsamples.

The regression results are presented in Table 7, Columns (7.1) and (7.2). We find that, for the variables of interest, *top reviews mean* and *signal reaffirmation*, coefficients for more popular products are either larger in magnitude, significant at stricter levels, or both, when compared to coefficients for less popular products. These findings hold when using our alternative measure of signal reaffirmation based on reviews textual content (see Table 7, Columns 7.4 and 7.5).

¹⁶ To derive the economic significance on *dispersion of subsequent ratings* from the main model, we derived the effect of *top reviews mean* (we multiplied its coefficient of -0.599 by a change of ratings of 0.25) and added it to the effect of *signal reaffirmation* (we multiply its coefficient of -0.476 by a change of 0.1).

¹⁷ As a robustness check, we adopted two alternate classifications for popularity. Rather than classifying products based on rank at the beginning of our data collection, our first alternate classification used the rank at the end of the data collection and the second used the average rank across the entire data collection, both using top 100 thresholds. Our findings are qualitatively similar.

Table 5. Dispersion of Subsequent Ratings Regression Results: Signal Reaffirmation

DV: Dispersion of subsequent ratings	Using mutual information (5.1)	Using textual content (5.2)
Top reviews mean	-0.599** (0.302)	-0.64** (0.29)
Signal reaffirmation	-0.476*** (0.201)	-0.61** (0.311)
All reviews mean	-0.649*** (0.245)	-0.557*** (0.229)
All reviews count	-1.43* (0.76)	-1.375** (0.655)
Signal consistency	-0.48*** (0.195)	-0.421* (0.234)
All reviews std. deviation	-0.287* (0.152)	-0.345*** (0.096)
Days since available	-0.093 (0.211)	-0.128 (0.262)
Sales rank	-0.034* (0.018)	-0.058*** (0.022)
Price	0.113 (0.15)	0.192 (0.162)
Ratio of rating extremes	1.131 (2.868)	1.141 (2.532)
Images posted	2.119** (1.028)	1.667 (1.499)
Videos posted	-0.367 (2.828)	-1.048 (3.285)
Amazon seller	-0.03 (0.038)	-0.029 (0.058)
Amazon shipper	0.062 (0.102)	0.103 (0.185)
DV lag	Y	Y
Product-fixed effects	Y	Y
Model fit statistics		
Wald χ^2	***	***
Prob. auto-regressive		
AR(1)	***	***
AR(2)	0.57	0.62
Hansen test of overidentification		
P	0.79	0.75
Panel	919	919
Observations	56,460	56,460

Note: Variables are derived for each product i and time t . Independent variables are lagged. DV lag is a lag of the dependent variable. Estimation is based on Equation (2) using AB/BB GMM. Signal reaffirmation is based on mutual information in Column (5.1) and is based on textual contents agreement in Column (5.2). Regression includes 77 instruments. Standard errors are listed in parentheses. Significance is *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table 6. Variable Definition and Descriptive Statistics Based on Product Popularity

Variable	More popular				Less popular			
	Mean	Std. dev.	Min	Max	Mean	Std. dev.	Min	Max
Dispersion of subsequent ratings	0.98	0.82	0	4	0.96	0.94	0	4
Top reviews mean	4.1	0.92	1	5	4.17	0.79	1	5
Signal reaffirmation	0.51	0.38	0	1.91	0.52	0.4	0	1.91
All reviews mean	4.4	0.43	1	5	4.34	0.54	1.2	5

All reviews count	6.1	1.35	0.69	9.94	5.63	1.17	0	9.94
Signal consistency	3.04	0.48	0	3.8	3.05	0.51	0	3.95
All reviews std. deviation	1.11	0.27	0	1.78	1.12	0.4	0	1.75
Days since available	7.31	0.68	4.6	8.65	7.25	0.73	4.34	8.65
Sales rank	3.92	1.3	0	7.52	6.19	2.31	0	14.11
Price	2.59	1.18	0	6.22	2.83	1.06	0	5.6
Ratio of rating extremes	-2.7	0.83	-5.3	0.33	-2.51	1.01	-5.88	3.23
Images posted	6.77	6.27	0	10	6.57	6.8	0	9.62
Videos posted	6.31	5.95	0	7.93	5.95	5.83	0	8
Amazon seller	0.38	0.49	0	1	0.35	0.48	0	1
Amazon shipper	0.58	0.49	0	1	0.8	0.4	0	1

Note: The *more popular* subsample has $N= 21,503$ observations with *Panels*= 350 products. The *less popular* subsample has $N= 34,957$ observations with *Panels*= 569 products.

Table 7. Dispersion of Subsequent Ratings Regression Results: More/Less Popular Products

DV: Dispersion of subsequent ratings	Using mutual information			Using textual contents		
	More popular (7.1)	Less popular (7.2)	All (7.3)	More popular (7.4)	Less popular (7.5)	All (7.6)
Top reviews mean	-0.409** (0.187)	-0.369** (0.17)	-0.778** (0.343)	-0.38** (0.169)	-0.418* (0.235)	-0.567** (0.261)
Signal reaffirmation	-0.448** (0.227)	-0.015 (0.351)	-0.81** (0.356)	-0.505** (0.227)	-0.065 (0.497)	-1.187** (0.576)
Lesspopular × Top Reviews Mean			0.199** (0.101)			0.282* (0.148)
Lesspopular × Sig Reaffirm			0.98* (0.567)			0.904 (0.903)
All reviews mean	-0.368** (0.185)	-0.343 (0.31)	-0.553* (0.291)	-0.206** (0.098)	-0.315 (0.197)	-0.529** (0.233)
All reviews count	-2.027** (0.937)	-1.046 (0.689)	-1.615** (0.751)	-2.226** (0.957)	-1.099 (0.96)	-1.655** (0.774)
Signal consistency	-0.333* (0.193)	-0.871 (0.609)	-0.748* (0.412)	-0.407* (0.219)	-0.93 (0.666)	-0.386 (0.431)
All reviews std. deviation	-0.136 (0.119)	-0.12 (0.153)	-0.3** (0.135)	-0.092 (0.07)	-0.135 (0.088)	-0.188*** (0.075)
Days since available	-0.27 (0.223)	-0.247 (0.204)	-0.041 (0.261)	-0.229 (0.243)	-0.279 (0.199)	-0.078 (0.207)
Sales rank	-0.099** (0.045)	-0.029 (0.083)	-0.16* (0.085)	-0.127* (0.075)	-0.021 (0.085)	-0.135* (0.075)
Price	0.013 (0.101)	0.233 (0.159)	0.105 (0.209)	0.001 (0.102)	0.235 (0.16)	0.067 (0.171)
Ratio of rating extremes	0.384 (1.655)	0.21 (2.526)	3.565 (3.223)	0.512 (1.734)	0.294 (2.59)	1.866 (2.589)
Images posted	-1.466 (1.653)	-0.703 (2.69)	-1.974 (1.74)	-1.574 (1.792)	-0.403 (2.64)	-2.384 (2.695)
Videos posted	-0.418 (1.82)	-0.362 (2.733)	-2.682 (3.415)	-0.305 (1.894)	-0.872 (2.802)	-1.059 (2.763)
Amazon seller	-0.064 (0.045)	0.069 (0.066)	-0.058 (0.064)	-0.059 (0.046)	-0.075 (0.07)	-0.019 (0.045)
Amazon shipper	0.053 (0.087)	0.041 (0.108)	0.09 (0.127)	0.041 (0.089)	0.03 (0.111)	0.066 (0.114)
DV lag	Y	Y	Y	Y	Y	Y
Product-fixed effects	Y	Y	Y	Y	Y	Y
Model fit statistics						
Wald χ^2	***	***	***	***	***	***

Prob. Auto-regressive						
AR(1)	***	***	***	***	***	***
AR(2)	0.45	0.63	0.45	0.48	0.55	0.53
Hansen test of overidentification						
<i>P</i>	0.25	0.21	0.31	0.32	0.17	0.26
Panel	350	569	919	350	569	919
Observations	21,503	34,957	56,460	21,503	34,957	56,460

Note: Variables are derived for each product *i* and time *t*. Independent variables are lagged. DV lag is a lag of the dependent variable. Estimation is based on Equation (2) using AB/BB GMM. Signal Reaffirmation is based on mutual information in Columns (7.1), (7.2) & (7.3), and is based on the textual contents' agreement in Columns (7.4), (7.5) & (7.6). Columns (7.1) and (7.4) show the results for more popular products; Columns (7.2) and (7.5) show the results for less popular products; Columns (7.3) and (7.6) show the results for all products. Regression includes 75 instruments. Standard errors are listed in parentheses. Significance is *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

We also supplement this analysis with the full sample while including interaction effects based on a dummy variable. The dummy, *LessPopular*, takes a value of 0 if a product sales rank is in the top 100, and 1 otherwise. For this alternative formulation, we modify Equation (2) to include the interaction effects between the *LessPopular* dummy variable and each of the two main signals, namely *top reviews mean* and *signal reaffirmation*. The interaction term with *top reviews mean* (*signal reaffirmation*) is used to evaluate the role of the signal from top reviews (*signal concordance*) when products are less popular.

The regression results from the interaction model are presented in Table 7, Column (7.3). The significant positive coefficients for the interaction term with *top reviews mean* (0.199 at $p < 0.05$ and 0.98 at $p < 0.1$) indicate that for less popular products, the impact of top reviews is diminished, resulting in a higher dispersion of opinions. As such, we find support for H4 that the impact of top reviews is weaker on the efficiency of information provisioning for less popular products compared to more popular ones. These findings are also supported when using the alternative measure of signal reaffirmation (see Table 7, Column 7.6). In a way, for these less popular products typically characterized by high variance (Sun, 2012), it is relatively more difficult to achieve concordance between top reviews and the overall set. Moreover, a higher variance introduces noise in the signals, be it the one coming from the overall mean or from top reviews, resulting in a relatively weaker impact of signals on the dispersion of reviews for less popular products.

Downstream Effect: Demand

We now build on our earlier analysis to explore the role that top reviews and related signals play in impacting the discriminant ability of customers in making purchases. Besides dispersion in terms of opinions, related research has identified information overload as a potential driver of online market failure. This is reflected in an uncertainty in product purchase decisions (Kim & Krishnan, 2015), which is eventually

translated into a reduction in demand. In this section, we therefore study the downstream implications of information overload as it relates to demand. In line with past research on online reviews, we use sales rank as a proxy for demand and investigate the downstream effect of information overload and the important role of top reviews. We adopt the same empirical approach, this time using *demand* (log of product sales rank) as the dependent variable. Because demand may be impacted by a platform-wide common shock affecting purchases from week to week, we include a weekly time dummy as well. Equation (3) is expressed as follows:

$$\begin{aligned} Demand_{i,t+1} = & \beta_2 Top\ reviews\ mean_{it} \\ & + \mu_2 Signal\ reaffirmation_{it} \\ & + \varphi_2 All\ reviews\ mean_{it} \\ & + \pi_1 All\ reviews\ count_{it} \\ & + \sum_c \delta_{c2} Control + \theta_2 Demand_{i,t} + \alpha_i \\ & + t + \epsilon_{it} \end{aligned}$$

where *i* refers to the product and *t* refers to the biweekly observation, *Demand_{i,t+1}* is the product sales rank at *t* + 1, *Top reviews mean_{it}* is the weighted mean of top reviews ratings, *Signal reaffirmation_{it}* is the mutual information between top reviews and all reviews ratings, *All reviews mean_{it}* is the mean of all reviews ratings, *All reviews count_{it}* is the total number of all reviews, **Control** is a vector of **c** control variables, α_i is a product-specific effect, *t* is a week indicator, and ϵ_{it} is the error term.

As previously, we supplement *all reviews mean* and *all reviews count* by controlling for *signal consistency*, *all reviews standard deviation*, *days since available*, *price*, *ratio of rating extremes*, *images posted*, *videos posted*, *Amazon seller* and *Amazon shipper*.

In Table 8 we present the results of our estimation of Equation (3) for *demand* (sales rank), showing the results for the subsample with more popular products only (Column 8.1), less popular products only (Column 8.2), and all products (Column 8.3).

Table 8. Demand Estimation Results: Signal Reaffirmation

DV: Demand	Using mutual information			Using textual contents		
	More popular (8.1)	Less popular (8.2)	All (8.3)	More popular (8.4)	Less popular (8.5)	All (8.6)
Top reviews mean	-0.88** (0.422)	-0.71 (0.507)	-1.711* (0.906)	-0.697** (0.353)	-0.338 (1.099)	-1.399* (0.808)
Signal reaffirmation	-0.575** (0.276)	-1.43 (5.649)	-1.288 (0.933)	-0.671** (0.341)	-1.993 (1.302)	-0.6 (1.173)
All reviews mean	-0.478* (0.285)	-1.05* (0.631)	-1.913*** (0.721)	-0.727* (0.438)	-1.507* (0.803)	-1.837** (0.929)
All reviews count	1.188* (0.703)	1.851 (1.655)	1.972* (0.997)	1.592* (0.946)	2.776 (2.554)	2.034* (1.214)
Signal consistency	-0.423 (0.586)	-0.509 (0.642)	-0.328 (1.447)	-0.524 (0.624)	-0.941 (1.039)	-0.378 (1.099)
All reviews std. deviation	-0.806 (1.002)	-1.825 (1.397)	-1.197 (0.972)	-0.898* (0.497)	-2.371** (1.092)	-2.79* (1.585)
Days since available	0.168 (1.016)	1.068 (2.417)	2.199 (1.008)	0.386 (1.943)	1.828 (2.788)	2.107 (2.053)
Price	-0.504 (0.478)	-0.806 (1.365)	-0.739 (1.344)	-0.84 (1.01)	-2.942 (2.594)	-0.925 (1.675)
Ratio of rating extremes	8.012 (7.012)	4.746 (7.149)	1.3 (3.866)	5.497 (5.855)	2.211 (5.611)	2.14 (5.815)
Images posted	-9.002 (7.358)	-2.445 (6.718)	-1.979 (4.336)	-6.579 (5.928)	-2.589 (5.922)	-1.754 (6.777)
Videos posted	7.931 (7.052)	5.353 (7.649)	1.635 (4.217)	5.438 (6.021)	2.249 (14.987)	3.421 (6.076)
Amazon seller	0.773 (1.837)	0.904 (2.353)	0.609 (2.819)	0.351 (2.006)	0.332 (4.474)	1.97 (3.614)
Amazon shipper	0.55 (1.521)	1.03 (1.059)	0.982 (0.686)	0.5 (1.826)	1.967 (1.444)	0.872 (0.663)
DV lag	Y	Y	Y	Y	Y	Y
Time-fixed effects	Y	Y	Y	Y	Y	Y
Product-fixed effects	Y	Y	Y	Y	Y	Y
Model fit statistics						
Wald χ^2	***	***	***	***	***	***
Prob. auto-regressive						
AR(1)	***	***	***	***	***	***
AR(2)	0.13	0.75	0.63	0.25	0.82	0.87
Hansen test of overidentification						
P	0.38	0.44	0.59	0.43	0.58	0.75
Panel	350	569	919	350	569	919
Observations	21,503	34,957	56,460	21,503	34,957	56,460

Note: Variables are derived for each product i and time t . Independent variables are lagged. DV lag is a lag of the dependent variable. Estimation is based on Equation (2) using AB/BB GMM. Signal Reaffirmation is based on mutual information in Columns (8.1), (8.2) & (8.3), and is based on the textual agreement in Columns (8.4), (8.5) & (8.6). Columns (8.1) and (8.4) show the results for more popular products; Columns (8.2) and (8.5) show the results for less popular products; Columns (8.3) and (8.6) show the results for all products. Regression includes 75 instruments. Standard errors are listed in parentheses. Significance is *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

We then reran all three estimations, this time using the alternative measure for signal reaffirmation, and present the results in Columns (8.4), (8.5), and (8.6). The coefficient in Column (8.3) for top reviews is significant and negative ($\beta_2 = -1.711$ at $p < 0.1$). This indicates that, as the signal from top reviews increases, the demand increases (lower sales rank) by 2.5% for a change in the values of variables from their 50th percentile to their 75th percentile. This finding affirms an economically significant downstream effect resulting from the efficiency of information provisioning. *Signal reaffirmation* is negative but not statistically significant, indicating a not terribly strong correlation of this signal with demand. It is also noticeable that less popular products are not impacted by these two signals (insignificant coefficients in Column 8.2). When exploring the impact for products that are more popular, the coefficients in Column (8.1) for top reviews is significant and negative ($\beta_2 = -0.880$ at $p < 0.05$) and for *signal reaffirmation* it is also significant and negative ($\mu_2 = -0.575$ at $p < 0.05$), indicating that, as the signal from top reviews and that from signal concordance increase, the demand increases for popular products, which is in line with the paper's main findings.

Conclusion

With the prevalent risk of information overload in online retailers' information-rich environments, our focus in this paper is on the role that signal selectivity, parsimony, and concordance play in shaping the efficiency of information provisioning. Our work investigates information signals selected from review-based sources that customers critically depend on, namely, the valence of a subset of reviews listed as *top reviews*. Such reviews are featured by the platform to arguably represent a somehow "distilled" signal from *all reviews*. The platform gathers reviews from customers, then gathers votes of approval on those reviews from other customers. It then features this selective subset of top reviews prominently for subsequent customers to consult. At the same time, customers still have access to all reviews, enabling them to assess the concordance of those reviews with the top reviews. We conceptualize the information signals as one derived from the selective valences of top reviews and another derived from the concordance between the distributions of top reviews and all reviews. We then study the role that top reviews play when (1) customers are overloaded with a large number of reviews, (2) the number of the top reviews themselves is not parsimonious, and (3) the signals

from the valences of top reviews and that from all the other reviews vary in their concordance. We characterize this setup as one of information provisioning, with a central role for the platform in this provisioning.

Using the lens of signaling in the presence of information overload, we are able to show the efficiency gains in information provisioning while analyzing a rich panel dataset from the largest online retailer in the U.S. The results indicate that top reviews are related to a non-negligible impact on the dispersion of subsequent ratings when the information available may cause overload. As such, we show that a rather simple tool, the featuring of top reviews, has major implications for online retailers. The impact of these valences of top reviews can be likened to the mind-swaying impact that celebrity endorsement introduces to decision-making. We also demonstrate the value of parsimony in the number of top reviews highlighted by a platform through a quasi-natural field experiment that we were fortunate to capture in our data collection. Exploiting this experimental setup, we show that the parsimony of top reviews further improves the efficiency of information provisioning. In particular, when only three reviews are presented as top reviews, compared to a higher number of reviews in the set, the impact is higher for incoming reviews.¹⁸ Additionally, the impact of signal concordance between top reviews and all reviews can be likened to the further mind-swaying that the additional endorsement of "the masses" brings. Our work zeroes in on a synergistic effect from such signal concordance and uses mutual information¹⁹ to contextualize it. We then further drill down into the differentiated relationships based on product popularity to identify instances where *signal reaffirmation* might break down and find that this to be indeed the case for less popular products.

Our findings contribute to the literature on signaling and on signal concordance. In fact, this study is the first attempt to conceptualize and demonstrate the importance of top reviews, their valence, their signal parsimony, and their signal concordance. We highlight the role that reaffirmation plays between the message derived from the valences of top reviews and those from all reviews and, more importantly, the efficiency of information provisioning. While we know that the way these signals are processed depends on the level of agreement with which they are received, empirical research in this area is still scarce. We therefore fill this gap by focusing on *signal concordance*, or the level of agreement between multiple signals, and study the varying implications of that signal concordance. This study is also the first to devise a way to

¹⁸ We note that three was the minimum number of top reviews that the online retailer we study displayed. As such, we did not observe a set with a smaller size such as one or two reviews. This leaves us with unanswered questions about further reducing the number of top reviews displayed and the impact of the resulting parsimony. Future research could explore the trade-off between the loss of information from parsimony and the ease of information processing.

¹⁹ We note that alternative measures of concordance that are seemingly simpler and straightforward may not capture the full extent of that concordance. One such measure is the difference between the mean of *top reviews* and that of *all reviews*. However, this difference between means can sometimes be positive and at other times be negative, making the interpretation of the findings very fuzzy. Additionally, this measure overlooks the variability in reviews.

operationalize the concordance between top reviews and all other reviews by developing a measure founded in information theory. This key novelty emulates the way that customers consume information, especially given the limited human ability to process information, and provides insights regarding underlying drivers of information efficiency.

Our findings also have design implications for practitioners. The focus on top reviews that platforms have adopted to overcome information overload is justified, especially when combined with some simple heuristics such as the histogram of reviews, which enables reaffirming the signal from top reviews. Practitioners should not expect to find a uniform impact on products, irrespective of their popularity level, but might have to explore further signals to improve information provisioning for less popular products. Additionally, the parsimony in presenting those top reviews is key, with the number being as small as three in our context.

Top reviews that are selected based on crowd endorsement and prominently displayed by retailers play a crucial role. While this practice has been lauded anecdotally for weakening the impact of manipulated reviews and the abundance of reviews, we provide concrete evidence of their value in strengthening review systems. Retailers should, however, pay particular attention to minimizing the cognitive dissonance of customers arising from the lack of concordance between top reviews and all reviews. Our results are also encouraging in terms of addressing the problem of review manipulation by producers or authors. Essentially, there is a great deal of emphasis on top reviews and their reaffirmation, and potentially less on individual reviews that might be fake. We should, however, caution that our empirical strategy does not allow us to analyze the counterfactual of not having any top reviews and the associated implications. Future studies might be interested in analyzing the role that the presence of top reviews plays in the informational landscape and comparing it to contexts in which such top reviews are completely missing.

We conclude with some implications for retailers and infomediaries. It is common practice for retailers to send follow-up emails to buyers requesting that they post a review of their purchase. Our research, however, highlights an alternate recourse for retailers. Given the impact of top reviews, an intuitive implication is for retailers to nudge customers, encouraging them to rate reviews' helpfulness. Retailers can start by actively seeking to build a critical mass of reviews and, once this is accomplished, they can turn to facilitate the crowd-endorsed process of elevating reviews into top reviews. In this way, rather than increasing the information overload of subsequent purchasers with more reviews, retailers can identify what they perceive to be useful reviews and motivate buyers to increase the visibility of these reviews by submitting helpfulness votes.

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Appendix A

Signal Reaffirmation

There are five events corresponding to observing 1, 2, 3, 4, and 5-star reviews, and there are five probabilities attached to those events. For all reviews, these probabilities can be readily derived from data using the ratio of the number of stars at this rating (i.e., 1, 2, 3, 4, or 5) and the total number of reviews. Additionally, these probabilities are intuitively available to customers through the histogram that online sellers display. These probabilities can also be readily derived for top reviews as the corresponding number of reviews is much smaller and thus can be derived with ease.

Computationally, deriving the *signal reaffirmation* requires both the joint distribution of ratings for top reviews and that of all reviews. As those joint distributions are not readily available, we rely on the corresponding marginals. It turns out that statistical theory and practice have been active areas of research over close to a century for imputing a joint distribution of variables based on their marginal distribution with wide-ranging implementations in physics, economics, engineering, risk analysis, and medicine (Lin et al., 2014) (see review by Balakrishnan and Lai (2009) and work by Gomez-Deniz et al. (2011) and more recently Cicalese et al., (2017)). Because of inherent computational challenges, however, only a handful of nonrestrictive methods have been successfully proposed for such imputation. In this paper, we follow the recent advancements made by Frogner and Poggio (2019) who define a general-purpose inference algorithm that is computationally efficient, applicable to real data, and superior in performance to existing approaches. We implement their approach (see algorithms 1 to 5 in their paper) to derive the joint distribution and then use this derivation to calculate MI between top reviews and all reviews for each product at time t and label it as *signal reaffirmation*. We note here that Algorithms 1-5 are based on the following theoretical derivation presented in Frogner and Poggio (2019):

The authors devise an algorithm that estimates the joint distribution of two categorical random variables $X: \Omega \rightarrow \{1, \dots, m\}$ and $Y: \Omega \rightarrow \{1, \dots, n\}$, given data from their marginal distributions, $\{x^{(k)}\}_{k=1}^{N_x} \sim P_X$, $\{y^{(l)}\}_{l=1}^{N_y} \sim P_Y$.

In their simplest case, the authors assume they are given full observations of the two marginal distributions: vectors $u \in \Delta^m$ and $v \in \Delta^n$ whose elements give the exact probabilities of the categories over which X and Y are defined. In this case, estimation entails finding a nonnegative matrix π satisfying marginal constraints $\pi \mathbf{1} = u$ and $\pi^T \mathbf{1} = v$.

The prior distribution comes from a general class of distributions, being those that are separable and log-concave of Legendre type, with support containing $\text{int}(\Delta^{m \times n})$.

Let $P_\pi(C)$ be a family of distributions over $\text{int}(\Delta^{m \times n})$, parameterized by $C \in \mathbb{R}$, and let $\Pr(\pi|C)$ denote the density with respect to the Lebesgue measure. The authors assume a common prior distribution $P_\pi(C)$ for N tables $\pi^{(i)}$, which satisfies the regularity properties. The prior $P_\pi(C)$ may be fit to such proxy data by maximum likelihood:

$$\hat{C} = \max_C \prod_{i=1}^N \Pr(\hat{\pi}^{(i)}|C), \text{ with } \hat{\pi}^{(i)} \text{ the estimated proxy for the joint distribution in the } i^{\text{th}} \text{ instance.}$$

Each observed vector defines an affine constraint on the table $\pi^{(i)}$; for marginals $u^{(i)}$ and $v^{(i)}$, the following is valid $\pi^{(i)} \mathbf{1} = u^{(i)}$ and $\pi^{(i)T} \mathbf{1} = v^{(i)}$. Together with the constraints that the entries of the table be nonnegative, these affine constraints define a convex polytope $U(u^{(i)}, v^{(i)})$ of tables consistent with the observations. Maximum a posteriori estimation reduces to finding a table $\pi_*^{(i)}$ that maximizes the prior over this polytope: $\pi_*^{(i)} = \underset{\pi \in U(u^{(i)}, v^{(i)})}{\operatorname{argmax}} \Pr(\pi|C)$. Using Bregman projection and Dykstra's method, the authors propose a series of algorithms (see algorithms 1 to 5 in their paper) to derive the joint distribution.

Appendix B

Topic Modeling and Sentiment Analysis

We used topic modeling and sentiment analysis to derive the measures for the variable *Signal Consistency* and for the alternative measure for the variable *signal reaffirmation*.

Derivation of the *signal consistency* variable: To assess the consistency of the textual content within top reviews, we relied on entropy as a well-established measure in information theory (Hall et al., 2008; Renyi, 1960; Shannon, 1951). Entropy has been used for measuring consistency in a variety of settings: different opinion groups (Dellarocas et al., 2007), numerical ratings (Li, 2018), software complexity (Harrison, 1992), and software portfolio (Temizkan et al., 2017). In general, entropy measures the amount of information (i.e., uncertainty) about an event associated with a given probability distribution. A higher value of entropy indicates greater uncertainty (in our case, inconsistency) in the text of the reviews. To quantify the entropy across the textual content of top reviews, we developed an approach that incorporates both the product dimensions discussed in the reviews (e.g., battery durability) as well as the sentiment expressed towards those dimensions, thus formulating dimension-sentiment (DS) pairs. The intuition is that as more combinations of DS pairs are mentioned, the entropy of the review system becomes larger indicating higher inconsistency (and less agreement) between top reviews. We conducted this analysis at the sentence level. This required identifying the review dimensions and sentiments, which we detail next.

Derivation of the *alternative signal reaffirmation* variable: To assess the consistency of the textual content between top reviews and all reviews, we quantified the overlap between the dimensions/opinions mentioned in top reviews and those mentioned in all reviews. To that end, we calculated the percentage of dimensions mentioned in top reviews among those mentioned in all reviews. We conducted this analysis at the review level. This required identifying the review dimensions, which we detail next.

Deriving review dimensions and sentiments: We applied topic modeling techniques to discover the review dimensions. This approach has been extensively used in a variety of settings (Blei et al., 2003; Griffiths & Steyvers, 2004; Hofmann, 1999). One such model is the latent Dirichlet allocation (LDA) (Blei et al., 2003), which assigns topics to documents in a corpus, based on the assumption that each such document is generated from a mixture of topics (which we label as dimensions), with each topic corresponding to a multinomial distribution over all the words in the documents. We conducted our analysis for *signal consistency* at the sentence level and applied LDA to derive the sentence-level topics (Buschken & Allenby, 2016) to assign a topic and a sentiment (positive or negative) for each review sentence. For *alternative signal reaffirmation*, we ran this analysis at the review level.

For the identification of dimensions in a category, and following standard implementation, we started by conducting tokenization, stopword removal, lemmatization, and stemming.²⁰ We first trained our LDA model at the review level to discover the topics.²¹ For each product category (Amazon labels the category as department which is also linked to the main product ranking), we formed a corpus of reviews of the products belonging to that category, which we used to conduct LDA topic selection.

Our initial approach was not to restrict the number of dimensions for any given category but rather to specify this number on the empirical results. To do so, we relied on a heuristic, *perplexity*, used as a convention in language modeling and computed on a hold-out test set (Blei et al., 2003). As this metric monotonically decreases in the likelihood of the test data, we started with 50 topics and reduce the number of dimensions until we obtained the smallest perplexity value which dictates the number of dimensions. For our dataset, the count of distinct topics ranged between 10 topics per category (the minimum we observe) and 15 (the maximum we observe). The LDA model also generated a list of keywords for each dimension as output. We used a list of 15 keywords per dimension and ask two researchers familiar with the context to ensure keywords coherence. As an additional validity step, we varied the iterations number in the derivation of LDA topics to check for any improvement in the coherence and meaningfulness of the keywords.

We ran the second step differently for *signal consistency* and *alternative signal reaffirmation*. For *alternative signal reaffirmation*, once the topics were identified, we calculated the ratio of the number of dimensions in the text of top reviews divided by the number of dimensions in *all reviews*. For *signal consistency*, once the topics were identified, we ran the trained LDA model on individual sentences to identify the topic (highest probability from LDA topics) that would be assigned to each sentence. A sample of topics and their corresponding keywords are provided in Table B1.

²⁰ For tokenization, we used the NLTK package. For stopwords, we used Genism's list in Stone et al. (2010) and supplemented it with nonrepresentative terms that occur in less than 2% of the reviews in our data set. For lemmatization, we used the WordNet lexical database of synonyms developed at Princeton University. For stemming, we used Porter stemmer initially published in Porter (1980).

²¹ This step overcomes issues from LDA running on short and sparse text in individual sentences.

Furthermore, in our context, the sentiment is essential in specifying the satisfaction corresponding to the dimension discussed. We therefore built on prior literature (also see Ordenes et al., 2017, for an extensive review) and used sentiment analysis to extract the sentiment of each of the dimensions (Jockers, 2017; Rinker, 2018). As such, we labeled each of the discussed dimensions in a review as having a (negative) positive sentiment to reflect the (dis)satisfaction with that dimension.

To derive the sentence-level sentiment, we used the *R* package “sentiment” (Rinker, 2018). This process compares each word in the document (here, sentence) with “positive” and “negative” dictionaries (Mohammad & Turney, 2010; Mohammad & Turney, 2013). Based on the level of matching between the words in the document and the dictionaries, the document is assigned a score ranging between -1 and 1. Using the resulting score, we assigned a positive polarity to scores higher than 0 and a negative polarity to scores lower than 0.

After dimensions and sentiments were identified, we used an entropy measure (Harris, 1982) to consolidate the DS pairs into our consistency metric that quantifies the dispersion of the informational content across future reviews. Derived this way, entropy is comparable across time periods, and is computed as follows. For product *i* at time *t*, the top reviews displayed consist of *P* sentences. Each sentence is assigned one of the *K* dimensions and one of two sentiments (positive or negative)—out of the $2 \times K$ possible dimension-sentiment pairs *DS*. Each DS pair has n_k sentences assigned to it with $\sum_1^{2K} n_k = P$. The proportion of sentences assigned to a given *DS* pair can be represented by n_k/P . Entropy is then expressed as $(\sum_1^{2K} n_k/P * \log(n_k/P))$ to measure the uncertainty in the content of the featured top reviews. We derive our main metric, *Signal Consistency* accordingly, by multiplying entropy by (-1) measured for each *i* at each *t*, which measures the consistency in the content of the featured top reviews. Thus, *Signal Consistency* is expressed as

$$\text{Signal Consistency}_{it} = \left(\sum_1^{2K} n_k/P * \log(n_k/P) \right)_{it}$$

Sample Text Mining Topics and Corresponding Keywords

In Table B1, we provide a sample result from our topic modeling. Using the category “cameras,” we show the 10 topics derived from LDA and the corresponding 10 keywords assigned for each of these topics.

Table B1. Topics Characterizing Category “Camera” and Their Corresponding Keywords									
Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8	Topic 9	Topic 10
camera	adult	default	econom	microsd	late	warranti	compact	build	screen
canon	child	usag	bore	connector	ship	repair	accept	bright	batteri
samsung	daughter	styl	kick	squar	excel	dust	weak	expand	pictur
app	father	board	execut	altern	arriv	waterproo	proper	strap	charg
devic	geek	blank	tile	rotat	work	reliabl	respond	angl	metal
buy	gift	seal	uniqu	chines	easi	durabl	self	mode	manual
product	girlfriend	cumberso	penni	code	year	heavier	starter	machin	research
play	grandkid	client	price	trash	time	refurbish	unrespons	larg	setup
game	husband	novic	pricey	slot	earlier	contact	malfunct	fair	icon
brand	xmas	devis	valu	storag	mail	servic	refus	mention	widget

Appendix C

Correlation Matrix

Table C1. Correlation Matrix for Dataset 1														
	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14
<i>Signal reaffirmation</i>	1													
<i>Signal consistency</i>	-0.018	1												
<i>All reviews mean</i>	-0.389	0.002	1											
<i>Top reviews mean</i>	-0.399	0.021	0.291	1										
<i>All reviews stddev</i>	0.372	-0.046	-0.607	-0.349	1									
<i>All reviews count</i>	0.034	0.104	0.172	0.034	-0.108	1								
<i>Days Since available</i>	-0.029	0.094	0.152	0.007	-0.082	0.539	1							
<i>Sales rank</i>	-0.040	-0.014	-0.345	0.034	0.348	-0.104	-0.05	1						
<i>Price</i>	0.001	0.112	0.406	0.004	-0.597	0.233	0.005	-0.248	1					
<i>Ratiorating extremes</i>	0.368	-0.052	-0.556	-0.329	0.460	0.188	0.099	0.326	-0.506	1				
<i>Images posted</i>	-0.045	0.097	0.412	0.083	-0.443	0.391	0.496	-0.252	0.382	-0.257	1			
<i>Videos posted</i>	0.118	-0.029	0.246	-0.037	-0.216	0.087	0.236	-0.212	0.229	-0.227	0.35	1		
<i>Amazon seller</i>	0.022	-0.148	-0.034	0.014	0.016	-0.273	-0.288	-0.103	-0.071	-0.093	-0.216	-0.002	1	
<i>Amazon shipper</i>	0.033	-0.089	-0.154	-0.044	0.166	-0.104	-0.016	-0.043	-0.306	0.127	-0.232	-0.051	0.525	1

Appendix D

Matching

Validating DID Assumptions

For both our dependent variables, *dispersion of subsequent ratings* and *demand*, we graphically checked the parallel trend prior to the start of the experimental setup. While there were level differences in Figure D1, the trend was quite similar before the treatment started. These give us confidence in the DID design.

In terms of the variation in the treated sample, in Figure D2, the x-axis shows the various time intervals for data collection. The y-axis shows the number of products. The depth shows the number of top reviews displayed (ranging from 4 to 10, once the experimental setup starts). The plot shows that for the treatment group there is a good variation in the number of top reviews. Thus, the varying treatment intensity aids with identification (Imbens & Wooldridge, 2007).

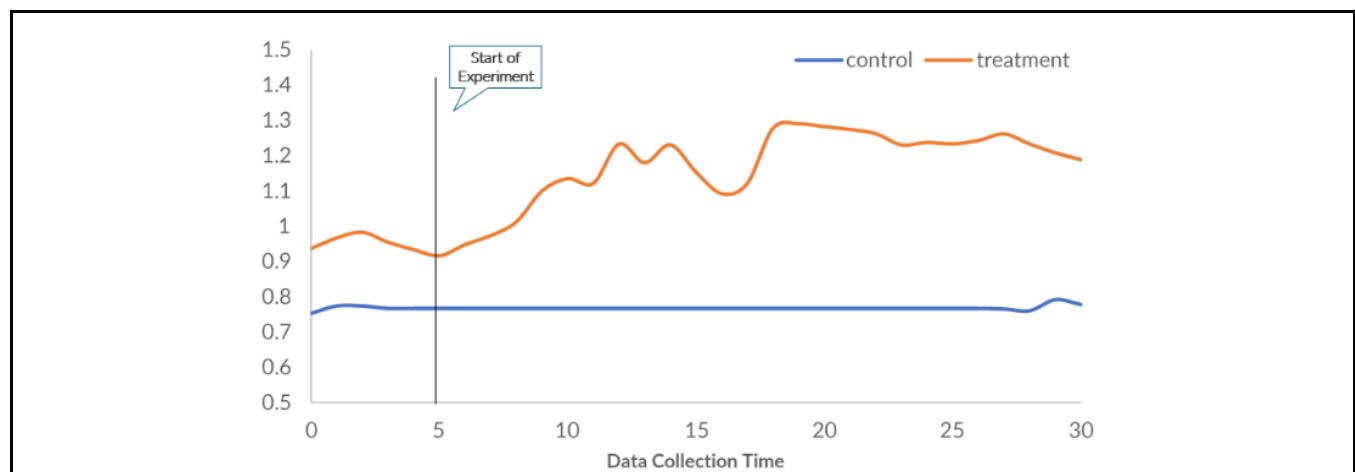


Figure D1. Parallel Trends for Dispersion of Subsequent Ratings.

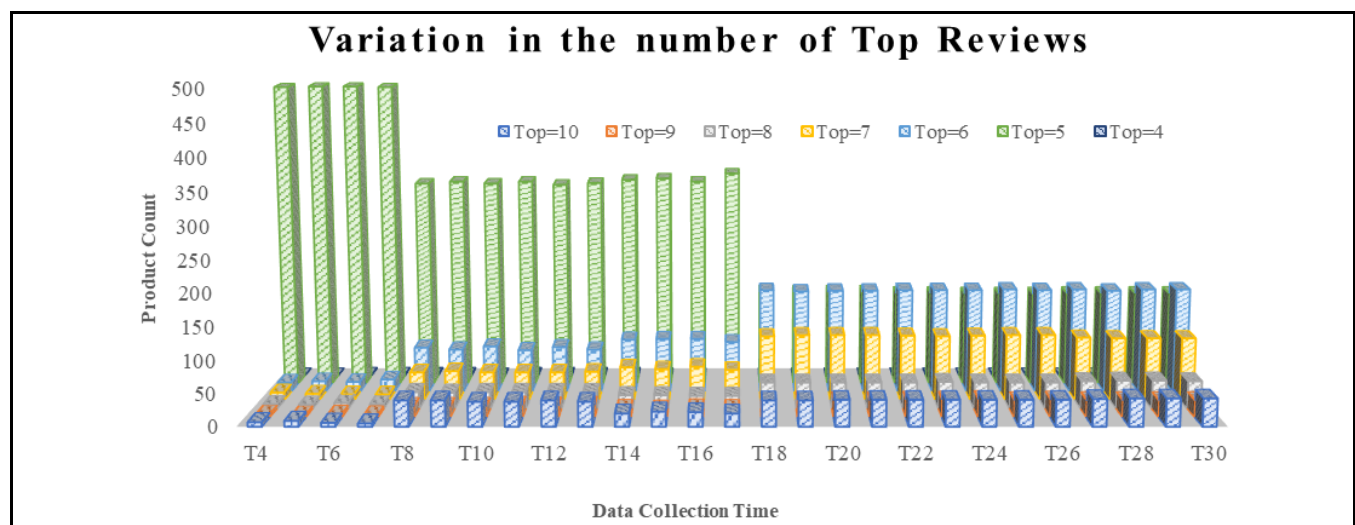


Figure D2. Variation in the Number of Displayed Top Reviews for the Treatment Group Ranging from 4 to 10

Summary Statistics for Variables Before and After Matching

The nearest neighbor (NN) matching between treated and control observations requires balancing the various variables used for matching (Hirano & Imbens, 2004). We therefore examine the balance between the treated and control products on different variables before and after matching. In this table, and for each of the variables used for matching, we show the mean before matching (Column 2 for the control group and Column 3 for the treatment) and after matching (Column 5 for the control group and Column 6 for the treatment group). We then report the *p*-value for the *t*-test of equal means (Columns 4 and 7).

As can be seen from this table, the *t*-test for mean differences (Column 4) is significant for most variables before matching, which indicates that the two groups of products differ on almost all variables. However, after NN matching, the mean differences of all the variables get closer to 0, with *p*-values that are not significant after matching (Column 7), indicating statistical balance and providing validity to our approach.

Table D1. Variable Means before and after Matching for Control and Treatment Groups						
	Unmatched			Matched		
	Mean		Sign.	Mean		Sign.
Variables	Control	Treatment	<i>p</i>-value	Control	Treatment	<i>p</i>-value
<i>Mean rating</i>	3.955	3.863	0.076	4.085	4.078	0.797
<i>Standard deviation rating</i>	1.305	1.216	0.039	1.2	1.201	0.918
<i>Review total</i>	5.536	5.409	0	5.519	5.505	0.265
<i>Days since available</i>	5.908	5.78	0.005	5.968	5.96	0.763
<i>Sales rank</i>	4.375	4.536	0	4.489	4.478	0.479
<i>Price</i>	5.316	5.588	0.016	5.492	5.469	0.853
<i>Ratio rating extremes</i>	-1.92	-1.831	0	-1.973	-1.94	0.086
<i>Images posted</i>	104.405	114.419	0.004	125.183	129.43	0.145
<i>Videos posted</i>	11.24	10.052	0.009	14.267	13.948	0.652
<i>Amazon seller</i>	0.304	0.311	0.559	0.304	0.311	0.559
<i>Amazon shipper</i>	0.628	0.642	0.04	0.629	0.628	0.372

Note: Sample size of 694 products

Appendix E

Difference-in-Differences Analysis for *Dispersion of Subsequent Ratings* with Alternative Matching Algorithms

To validate our quasi-experimental setup, we conducted matching between control and treatment units using alternative matching algorithms. Because each matching algorithm is subject to some limitations, replicating the analysis with a variety of algorithms ensures confidence in our findings. We started by implementing inverse propensity weighting (IPW) (Guo et al., 2017). The IPW estimator was derived by weighting each treated observation's outcome in the sample by the inverse of their estimated probability of receiving treatment (the propensity score), and weighting each untreated observation's outcome by the reciprocal of one minus their propensity score. IPW was used to weight each observation in the treated group, and one minus the inverse of the propensity score (i.e., the propensity of not being in the treated group) in the controls and has been shown to produce unbiased estimates (Hirano & Imbens, 2001; Imbens, 2000). We also applied coarsened exact matching (CEM) (Burtch et al., 2018; Leung & Sharkey, 2014) which mitigates the problem of not finding matches by coarsening the observed variables and performing an exact match on the coarsened data, generating matches with lower estimation error compared to other methods.

Estimation results of the DID model with categorical treatment implementing IPW (inverse propensity weighting) and CEM (coarsened exact matching) as alternative matching algorithms are presented in Table E1 for the *dispersion of subsequent ratings*. The findings from both regressions with alternative matching algorithms are consistent with our findings using NN matching, further confirming that parsimony is more effective for the efficiency of information provisioning.

Table E1. *Dispersion of Subsequent Ratings* Regression Results: DID Model with Categorical Treatment using IPW and CEM Matching

<i>DV: Dispersion of subsequent ratings</i>	IPW	CEM
	(1)	(2)
<i>Treatment (categorical)</i>	0.039** (0.017)	0.255*** (0.078)
<i>Top reviews mean</i>	-0.339** (0.159)	-0.702** (0.307)
<i>Signal reaffirmation</i>	-0.469** (0.21)	-0.565** (0.265)
<i>All reviews mean</i>	-0.477* (0.245)	-1.877* (0.987)
<i>All reviews count</i>	-0.037 (0.073)	-0.408 (2.567)
<i>Signal consistency</i>	-0.371** (0.166)	-2.791** (1.398)
<i>All reviews std. deviation</i>	-2.224 (1.667)	-3.455 (3.079)
<i>Days since available</i>	-0.072 (0.067)	-0.018 (2.555)
<i>Sales rank</i>	-0.032** (0.016)	-0.271** (0.127)
<i>Price</i>	0.149* (0.081)	0.061* (0.034)
<i>Ratio of rating extremes</i>	0.456 (0.326)	3.667** (1.649)
<i>Images posted</i>	0.193** (0.094)	1.063 (1.05)
<i>Videos posted</i>	-0.021 (0.024)	-0.03 (0.024)
<i>Amazon seller</i>	-0.041* (0.022)	-0.287* (0.156)
<i>Amazon shipper</i>	0.007 (0.02)	0.133* (0.076)

Time fixed-effects	Y	Y
Product fixed-effects	Y	Y
Model fit statistics		
R^2	0.27	0.43
Panel	596	596
Observations	14,857	14,857

Note: Variables are derived for product i and time t . Independent variables are lagged. The treatment is a categorical variable. Estimation uses robust OLS with product- and time-fixed effects. Matching is based on IPW in Column (1) and CEM in Column (2). Standard errors are in parentheses. Significance is *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Appendix F

Random Implementation Analysis

Spurious relationships or serial correlation in the dependent variables may potentially result in false significance in our estimates. Also, because the data we collected was spread over several months, it may have been possible that idiosyncrasies about the platform and the products were driving the observed effect. To address this problem, we conducted a random implementation test (Bertrand et al., 2004) randomly reassigning the treatment indicator in our data. That is, we shuffled the treatment indicators in the dataset to a randomly selected set of control observations, thereby creating a placebo treatment. We then estimated our standard difference-in-differences model and stored the coefficients of this placebo treatment. We replicated this procedure 1,000 times to assess whether the placebo treatment was also significant, which would have resulted from spuriousness caused by autocorrelation in the dependent variable (Abadie et al., 2010; Bertrand et al., 2004; Burtch et al., 2018). Figure F1 displays the distribution of the resulting values of the placebo treatment coefficient for *dispersion of subsequent ratings*. The distribution is tightly centered around zero, thus indicating that, had the treatment been randomly assigned, the resulting impact of *top reviews* on the *dispersion of subsequent ratings* would be insignificant.

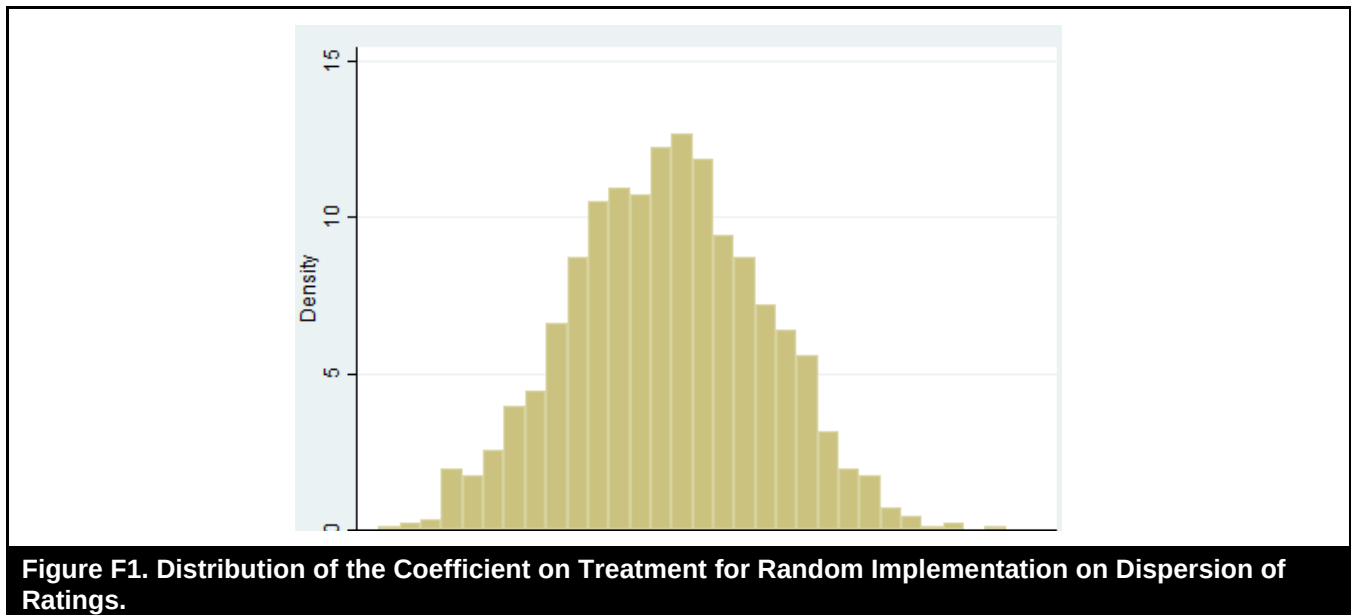


Figure F1. Distribution of the Coefficient on Treatment for Random Implementation on Dispersion of Ratings.

Appendix G

Main Analyses using Dataset 2

We repeated the paper's main analysis (Equation 2) using Dataset 2 to ensure that our findings are qualitatively identical. We provide the estimation results in Table G1. Reassuringly, the results are qualitatively similar. We followed identical preprocessing steps as for Dataset 1. This included dropping duplicate product listings and products with missing data such as sales rank or price. It also included ensuring that all the products had a sales rank reported at the category level and had five or more reviews each time interval.

Table G1. Dispersion of Subsequent Ratings Regression Results with Dataset 2	
Dispersion of subsequent ratings	
Top reviews mean	-0.844** (0.386)
Signal reaffirmation	-0.904* (0.5)
All reviews mean	-0.481** (0.223)
All reviews count	-1.057* (0.549)
Signal consistency	-0.287* (0.149)
All reviews std. deviation	-0.523 (0.398)
Days since available	-0.332 (0.241)
Sales rank	-0.116 (0.136)
Price	0.05 (0.238)
Ratio of rating extremes	0.793 (0.538)
Images posted	0.303** (0.146)
Videos posted	-0.193 (0.615)
Amazon seller	-0.114** (0.054)
Amazon shipper	0.118 (0.107)
DV lag	Y
Product FE	Y
Model fit statistics	
Wald χ^2	***
Prob. auto-regressive	
AR(1)	***
AR(2)	0.78
Hansen test of overidentification	
p	0.84
Panel	596
Observations	14,857

Note: Variables are derived for each product i and time t . Independent variables are lagged. DV lag is a lag of the dependent variable. Estimation is based on Equation 3 using AB/BB GMM. Regression includes 80 instruments. Standard errors are listed in parentheses. Significance is *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.